MIL-M-19097B(SHIPS)
22 May 1961
SUPERSEDING
MIL-M-19097A(SHIPS)
1 March 1956

MILITARY SPECIFICATION

MOTOR-GENERATORS, DC TO AC, SHIPBOARD SERVICE

1. SCOPE

1.1 Scope. - This specification covers direct current (d. c.) to alternating current (a. c.) motor generator power supplies for Naval shipboard use.

1.2 Classification. - Motor generators and associated control equipment shall be of the following types as specified (see 6. 1):

Type I - General-purpose-intended for surface ships.
Type II - Special-intended for submarine use only.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearing, Ball, Annular, (General Purpose).
QQ-I-666 - Iron Castings, Malleable.
GGG-P-781 - Puller and Puller Kit, Mechanical and Mechanical Puller Attachment.

MILITARY

MIL-E-1 - Electron Tubes and Crystal Rectifiers.
MIL-R-22 - Resistor, Variable (Wirewound, Power Type).
MIL-C-25 - Capacitors, Fixed, Paper-Dielectric, Direct-Current Hermetically Sealed in Metallic Cases, General Specification For.
MIL-C-62 - Capacitors, Fixed Electrolytic DC, Aluminum Dry Electrolytic, Polorized).
MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-S-901 - Shockproof-Equipment, Class HI (High Impact), Shipboard Application, Test for.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-963 - Drawings; Production, Procedure For Procurement Of.
MIL-I-1361 - Instruments, Electrical Measuring: Shunts, Resistors, and Transformers (Ship and Shore Use).
MIL-C-2174 - Controllers, Motor Starters, and Master Switches—Direct-Current, Naval Shipboard Use.
MIL-G-3545 - Grease, Aircraft, High Temperature.
MIL-B-3743 - Brushes Electrical, Contact; and Carbon Stock, Electrical Contact Brush.
MIL-C-5015 - Connectors, Electrical AN Type.
MIL-Q-9858 - Quality Control System Requirements.
MIL-M-10304 - Meters, Electrical Indicating, Panel Type, Ruggedized.
MIL-M-15071 - Manual, Technical, for Mechanical and Electrical Equipment, Less Electronics.
MIL-S-15083 - Steel; Casting.
MIL-R-15109 - Resistors and Rheostats, Shockproof (Naval Shipboard Use).
MIL-S-15291 - Switches, Rotary, Snap Action, Bureau-Design.
MIL-L-15719 - Lubricating Grease (High-Temperature, Electric Motor, Ball and Roller Bearings).
MIL-P-16018 - Packaging of Switchboards, Controllers and Motor Starters (Over 40 Pounds).
MIL-M-16034 - Meters, Electrical-Indicating (Switchboard and Portable Types).
MIL-I-16104 - Indicators, Synchronization.

FSC 6125

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MILITARY (CONT'D)

MIL-M-16125 - Meters, Electrical Frequency.
MIL-P-16298 - Preservation, Packaging, Packing, and Marking of Electric Machines Having Rotating Parts (and Associated Repair Parts).
MIL-I-17166 - Iron Castings; Nodular Graphitic (Ductile Iron) (for Shipboard Applications).
MIL-M-17191 - Mounts, Resilient; Portsmouth Bonded Spool Type.
MIL-C-17361 - Circuit Breakers, Air, Electric, Insulated Enclosure (Shipboard Use).
MIL-M-17508 - Mounts, Resilient, Naval Engineering Experiment Station Types 6E2000, 6E900, 7E450, 6E150 and 6E100.
MIL-E-17555 - Electronic and Electrical Equipment And Associated Repair Parts; Preparation For Delivery Of.
MIL-C-17568 - Circuit Breaker (Automatic ALB) And Switch, Toggle (Circuit Breaker, Non-Automatic - NLB), Air, Insulated Enclosure, 125 Volts And Below, AC Or DC, Naval Shipboard.
MIL-B-17931 - Bearings, Ball Annular Specially Selected for Quiet Operation, Noise Tested.
MIL-R-18261 - Rectifiers, Metallic; Selenium, Copper Oxide, and Magnesium-Copper Sulfide Types, Naval Shipboard Use.
MIL-M-19379 - Mounts, Resilient, Mare Island Rubber Laboratory Types 11M25, 11M15 And 10M50.
MIL-S-19500 - Semiconductor Devices, General Specification for.
MIL-E-22843 - Equipment, Low Noise Level (Naval Shipboard).

STANDARDS

MILITARY

MIL-STD-167 - Mechanical Vibration of Shipboard Equipment.
MIL-STD-195 - Marking of Connections for Electric Assemblies.
MIL-STD-242 - Electronic Equipment Parts (Selected Standards).
MIL-STD-701 - Preferred And Guidance List Of Transistors.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Ave. at 33rd St., N.Y. 16, N.Y.).

3. REQUIREMENTS

3.1 Definitions

3.1.1 Rated condition. - Rated condition is characteristics assigned to a machine such as, speed, voltage, frequency and current for a specified output rating.

3.1.2 Nominal rating. - Nominal rating is the normal constant operating condition or characteristic assigned to a machine when a range of operating conditions or characteristics are specified.

3.1.3 Continuous duty. - Continuous duty is a requirement of service that the equipment shall operate satisfactory at any constant load in the range of no-load to full load for an indefinitely long time.

3.1.4 Total harmonic content. - The total harmonic content is the total root mean square (r.m.s.) voltage remaining when the fundamental component is removed.

3.1.5 Voltage modulation. - Voltage modulation is the cyclic variation or random dynamic variations, or both, about an average of the ac peak voltage during steady state electric-system operations such as caused by voltage regulation and speed variations. The modulation envelope is formed by a continuous curve connecting each sine wave peak to the successive sine wave peak.

3.1.6 Short-circuit ratio. - The short-circuit ratio is the ratio of the field current at rated open circuit armature voltage and frequency to the field current required to produce rated armature current for a sustained symmetrical short circuit at rated frequency.

3.1.7 Synchronous impedance. - The per unit direct axis synchronous impedance equals the ratio

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of the field current at rated armature current on sustained symmetrical short circuit to the field current at normal open circuit voltage on the air gap line.

3.1.8 Air-gap line. - The air-gap line is the extended straight line part of the no-load saturation curve.

3.1.9 Stationary air-gap. - Stationary air-gap is the air gap measured between the center of each pole and the armature when the rotating elements are stationary.

3.2 Performance requirements. -

3.2.1 Rating. - The motor generators shall be furnished in the following kilowatt (kw) ratings as specified (see 6.1).

(a) Type I - 1.5, 3, 5, 10, 15, 30, 60 and 100 KW.
(b) Type II - As specified.

3.2.2 Input voltage. - The motor generators and associated control equipment shall be designed to operate on one of the voltages in table I as specified (see 6.1.).

Table I - Input voltages.

Type for motor generator	Nominal voltage	Voltage range min.	max.
Type I	115	105	125
Type I	230	210	250
Type II	250	210	355

3.2.3 Output voltage. - The output voltage of the motor generator shall be either 120 or 450 volts as specified (see 6.1).

3.2.4 Phase. - The phase of the output voltage shall be either single phase or three phase as specified (see 6.1).

3.2.5 Frequency. - The output frequency of the motor generator shall be either 60 or 400 cycles as specified (see 6.1).

3.2.6 Power factor. - The power factor of 0.8 shall be used.

3.2.7 Duty. - Motor generators and control equipment shall be designed for continuous duty when operating on the input voltage range specified in 3.2.2.

3.2.8 Ambient temperature. - The motor generators and all associated control equipment shall be designed for operation in an ambient temperature of 50° Centigrade (C.).

3.2.9 Shock. - Motor generators and all associated control equipment shall be capable of withstanding the high impact shock test as specified for Grade I equipment of Specification MIL-S-901.

3.2.10 Vibration (type II motor generators only). - All parts and the complete structure of the motor generator and control equipment shall be designed to withstand externally excited vibration type I as specified Standard MIL-STD-167.

3.2.11 Radio interference. - The motor generators with associated control equipment shall be designed to produce minimum conducted radio interference within the frequency range of 150 kilocycles (kc) to 20 megacycles (mc). Use of filters or condensers to reduce radio interference shall be permitted in control equipment only. As means of reducing radio interference special attention should be given to design features specified in 3.4.22.

3.2.12 Grounding. - Equipment shall be designed to operate satisfactorily from an ungrounded ship's power system but accidental grounding of this system shall not cause mal-operation. Design and construction shall be such that all exposed metal parts are at ground (ship's hull) potential at all times. None of the circuits of the equipment furnished shall be tied to ground. The output of the motor generator power supply shall also be designed as an ungrounded system; however, the equipment shall be designed to perform in accordance with the requirements of this specification in the event of single failure to ground and when operating with loads which may have high impedance circuits to ground such as may be caused by radio interference suppression networks.

3.2.13 Modes of operation. - The equipment shall be suitable for operation in any of the following manners specified herein:

(a) Voltage regulator and frequency regulator in automatic.
(b) Voltage regulator and frequency regulator in manual.
(c) Voltage regulator in automatic, frequency regulator in manual.
(d) Voltage regulator in manual, frequency regulator in automatic.

3.2.14 Voltage regulation. - Voltage regulation shall be achieved by a completely static regulator and excitation system. The voltage regulator shall be capable of maintaining the voltage or mean voltages for 3 phase generators within plus or minus 0.5 percent of rated voltage under the following conditions or any combination thereof:

(a) Load varying between no load and rated load.

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(b) Load power factor varying between 0.8 lagging and unity.
(c) Variation of generator frequency within plus or minus 2 percent of rated frequency.
(d) Variation of input voltage within the range specified.
(e) Variation of equipment temperature from cold at specified ambient to full load rise at specified ambient.
(f) Unbalanced generator load of 15 percent of rated output (for 3 phase generators).
(g) Generator voltage adjusted to any point within the regulator band.

3.2.15 Voltage deviation and recovery. - The sudden application of a 50 percent load at 0.8 factor lagging to the output of the generator shall result in a voltage dip not greater than 9 percent of rated voltage. The time to return to and stay within plus or minus 0.5 percent of the initial value of voltage shall not exceed 0.5 second for 400 cycle generators and one second for 60 cycle generators. This value of recovery time and voltage deviation shall be met also upon removal of 50 percent 0.8 power factor load in one step from 100 percent load.

3.2.16 Frequency regulation. - Frequency regulation shall be achieved by the use of a completely static frequency regulator. The frequency regulator shall maintain the output frequency to within plus or minus 0.5 percent of rated frequency under the following conditions or any combination thereof:

(a) Load varying between no load and rated load.
(b) Load power factor varying between 0.8 lagging and unity.
(c) Variation of input voltage within the range specified.
(d) Variation of equipment temperature from cold at specified ambient to full load rise at specified ambient.
(e) Variation in generator voltage as specified in 3.2.18.
(f) Generator frequency adjusted to any point within the regulation band.

3.2.17 Frequency deviation and recovery. - Upon application of 50 percent 0.8 power factor load from no load in one step, and with rated voltage applied to the motor, the frequency shall recover to within 1/2 percent of rated value within 0.5 second for 400 cycle generators and one second for 60 cycle generators. Under these conditions the maximum frequency deviation shall not exceed 4 percent of the initial value of frequency. This value of recovery time and frequency deviation shall be met also upon removal of 50 percent 0.8 power factor load in one step from 100 percent load.

3.2.18 Regulated voltage adjustment. - Provisions shall be made to adjust the regulated voltage within the range of 5.0 percent above and below rated volts in steps of not more than 0.2 percent of rated voltage at any load from no load to 100 percent of rated load at rated power factor.

3.2.19 Manual voltage adjustment. - Provision shall be made for manual control of voltage within the limits specified in 3.2.18.

3.2.20 Regulated frequency adjustment. - Provisions shall be made to adjust the regulated frequency within the range of plus or minus 2 percent of rated frequency in steps of not more than 0.1 percent at any load from no load to full load at rated power factor and with any voltage within the specified range applied to the motor.

3.2.21 Manual frequency adjustment. - Provision shall be made for manual control of frequency within the limits specified in 3.2.20.

3.2.22 Harmonic content. - From no load to rated load, the r.m.s. value of the sum or single harmonic component in the output wave shall not exceed the percentages of the r.m.s. value of the rated output voltage as specified in table II.

Table II - Harmonic content.

Frequency	Sum of harmonics	Single harmonic
cycle	percent	percent
60	5	3
400	3	2

3.2.23 Amplitude modulation. - Amplitude modulation of the output voltage of the generator when operating with its intended voltage regulator shall not exceed 0.5 percent of the peak voltage. This requirement shall apply for any loads between no load and rated load and at any power factor between 0.8 (lagging) and unity.

3.2.24 Line unbalance (three phase generators only). - The variation between line voltages shall not be greater than 5 percent of rated voltage when a single phase load of 15 percent rated current at any power factor between 0.8 and unity is applied to any line of the generator which is already loaded with a balanced 3 phase load of 75 percent of rated current.

3.2.25 Voltage balance (three phase generators only). - The variation between line voltages at the output terminals of the power supply (load side of circuit breaker) shall not be greater than 0.5 percent at no load or for any balanced three phase current up to rated current at any power factor between 0.8 and unity.

3.2.26 Generator short circuit ratio. - The generator short circuit ratio shall be no less than 0.8.

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3.2.27 System short circuit performance.- The static exciter and voltage regulator shall be capable of producing sufficient generator field current to maintain 300 percent rated generator output current for a period of 2 minutes under the following 3-phase short circuit condition for 3 phase generators or one phase short circuit condition for one phase generators:

Initial load	Control	Generator field temperature	Symmetry
No load	Automatic	Hot	Symmetrical

Hot field temperature shall be equivalent to normal full load rise obtained from the heat run plus specified ambient temperature (50°C.). The design of the motor generators and control equipment shall be such that it can withstand a short circuit current of the magnitude required for 2 minutes without damaging the equipment. Upon removal of the short circuit, rated speed and voltage shall be automatically restored.

3.2.28 Parallel operation.- Parallel operation shall be provided as follows:

(a) The equipment shall be designed so that any number of motor generators and control equipment of identical design will operate in parallel in accordance with the requirements specified herein. Parallel operation shall be accomplished by cross-current compensation of the voltage regulators and speed governors rather than by droop compensation.

3.2.28.1 Parallel operation performance.- The maximum difference in kilovar (kvar) and kw supplied by generators operating in parallel shall not exceed the following limits under any load condition between 20 and 100 percent of rated load. These requirements shall also apply for input voltage variations specified in table I, variations in equipment temperature from ambient to full load rise, and variation in power factor between 0.8 lagging and unity. The design shall be such that the equipment will perform as specified herein when each power supply is connected to a different source and without any electrical interconnection between these sources. The requirements shall apply only when both the speed and voltage regulators are in automatic operation.

(a) The maximum difference in kw supplied by the generators in parallel shall not exceed 8 percent of the kw rating of one of the generators, for 115 and 230 nominal input voltage and 10 percent for 250 nominal input voltage.

(b) The maximum difference kvar supplied by the generators in parallel shall not exceed 6 percent of kvar rating of one of the generators.

3.2.29 Insulation resistance.- The insulation resistance, when corrected in accordance with 4.6.6 shall not be less than 50 megohms for motor generators and 10 megohms for associated controls.

3.2.30 Circuit segregation.- The d.c. input and a.c. output shall be completely isolated at all times.

3.2.31 Dielectric strength.- The motor generator and associated control equipment shall be capable of withstanding a dielectric strength test (see 4.6.7) of twice the normal voltage of the circuit plus 1,000 volts for one minute.

3.3 Detailed requirements-

3.3.1 General.- Equipment covered by this specification shall conform to Specification MIL-E-917 unless otherwise specified herein. The equipment shall form a complete power supply consisting of a motor generator and its intended control equipment.

3.3.2 Reliability.- To meet the needs of the Naval service, it is imperative that reliability of operation be considered of prime importance in the design and manufacture of the equipment. The contractor shall employ all reasonable methods possible in the process of manufacture to assure quality and maximum reliability consistent with the state of the art. In the functional application of parts to equipment circuits, adequate factors of safety shall be provided by suitable deviations from their specification values where required, in order to insure high equipment reliability under all service conditions. The design shall include all possible features which will result in reliable and stable operation with minimum need for adjustment and alignment, minimum frequency of failure and minimum requirements for maintenance. Maintenance shall be simplified to reduce the need for highly skilled maintenance personnel. The final product shall reflect the utmost in simplicity, have maximum reliability and be easy to install and maintain.

3.3.3 Interchangeability.- For this equipment, all essential parts including removable units, repair parts, and other items, shall be interchangeable physically and electrically with corresponding items or parts without selective matching, drilling, filing, or the use of undue force. Readjustment and retuning of the regulators will be permitted after components are replaced. Interchangeability of repair parts is necessary to permit both ship and shore maintenance

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Initial load	Control	Generator field temperature	Symmetry
No load	Automatic	Hot	Symmetrical

Hot field temperature shall be equivalent to normal full load rise obtained from the heat run plus specified ambient temperature (50°C.). The design of the motor generators and control equipment shall be such that it can withstand a short circuit current of the magnitude required for 2 minutes without damaging the equipment. Upon removal of the short circuit, rated speed and voltage shall be automatically restored.

3.2.28 Parallel operation.- Parallel operation shall be provided as follows:

(a) The equipment shall be designed so that any number of motor generators and control equipment of identical design will operate in parallel in accordance with the requirements specified herein. Parallel operation shall be accomplished by cross-current compensation of the voltage regulators and speed governors rather than by droop compensation.

3.2.28.1 Parallel operation performance.- The maximum difference in kilovar (kvar) and kw supplied by generators operating in parallel shall not exceed the following limits under any load condition between 20 and 100 percent of rated load. These requirements shall also apply for input voltage variations specified in table I, variations in equipment temperature from ambient to full load rise, and variation in power factor between 0.8 lagging and unity. The design shall be such that the equipment will perform as specified herein when each power supply is connected to a different source and without any electrical interconnection between these sources. The requirements shall apply only when both the speed and voltage regulators are in automatic operation.

(a) The maximum difference in kw supplied by the generators in parallel shall not exceed 8 percent of the kw rating of one of the generators, for 115 and 230 nominal input voltage and 10 percent for 250 nominal input voltage.

(b) The maximum difference kvar supplied by the generators in parallel shall not exceed 6 percent of kvar rating of one of the generators.

3.2.29 Insulation resistance.- The insulation resistance, when corrected in accordance with 4.6.6 shall not be less than 50 megohms for motor generators and 10 megohms for associated controls.

3.2.30 Circuit segregation.- The d.c. input and a.c. output shall be completely isolated at all times.

3.2.31 Dielectric strength.- The motor generator and associated control equipment shall be capable of withstanding a dielectric strength test (see 4.6.7) of twice the normal voltage of the circuit plus 1,000 volts for one minute.

3.3 Detailed requirements-

3.3.1 General.- Equipment covered by this specification shall conform to Specification MIL-E-917 unless otherwise specified herein. The equipment shall form a complete power supply consisting of a motor generator and its intended control equipment.

3.3.2 Reliability.- To meet the needs of the Naval service, it is imperative that reliability of operation be considered of prime importance in the design and manufacture of the equipment. The contractor shall employ all reasonable methods possible in the process of manufacture to assure quality and maximum reliability consistent with the state of the art. In the functional application of parts to equipment circuits, adequate factors of safety shall be provided by suitable deviations from their specification values where required, in order to insure high equipment reliability under all service conditions. The design shall include all possible features which will result in reliable and stable operation with minimum need for adjustment and alignment, minimum frequency of failure and minimum requirements for maintenance. Maintenance shall be simplified to reduce the need for highly skilled maintenance personnel. The final product shall reflect the utmost in simplicity, have maximum reliability and be easy to install and maintain.

3.3.3 Interchangeability.- For this equipment, all essential parts including removable units, repair parts, and other items, shall be interchangeable physically and electrically with corresponding items or parts without selective matching, drilling, filing, or the use of undue force. Readjustment and retuning of the regulators will be permitted after components are replaced. Interchangeability of repair parts is necessary to permit both ship and shore maintenance

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Table III - Maximum permissible temperature rise (degrees C)
(based on 50°C Ambient - continuous rated load)

Name of part (3)	Insulation class			
	B	F	H	C (1)
Bearings	60	60	100	
Coils -				
Measured by thermocouples or thermometers	60	85	120	
Measured by resistance	70	95	140	
Cores and mechanical parts adjacent to insulation	60	85	120	
Commutators & collector rings	75	100	135	
Air-within 1/2 inch of meters	15	15	15	
Capacitors - Air within 1/2 inch of case	35	35	35	
Rectifiers - selenium (2)	25	25	25	
Rheostats & resistors:				
bare resistor material	375	375	375	
resistor embedding material	300	300	300	
Bolted connections and terminal studs:				
Not plated	50	50	50	
Silver plated	65	65	65	
Contact:				
Solid	70	70	70	
Solid, silver faced	80	80	80	

(1) Maximum temperature depends on type of insulation used.
(2) Silicon rectifiers, voltage reference elements, voltage regulator elements and transistors - permissible temperature rise on these parts in dependent upon operating conditions. Maximum permissible case temperature shall be shown on the equipment drawings. (see 3.7.32 (6)(a)).
(3) Where no measurement method is indicated, thermometer or thermocouple method is used.

shall be in accordance with Specification MIL-E-22843 and the following:

(a) Type I motor generators (rotating equipment only)
Grade - Grade C equipment.
Operating conditions - No load to full load at the specified input voltage.

(b) Type II motor generators and associated controls
Grade - Grade A or B as specified (see 6.1).
Operating conditions - No load to full load over specified input voltage range.

3.3.13 Structureborne noise limits for type II motor generators and associated controls. - The structureborne noise limits and measurements shall be in accordance with Specification MIL-E-22843 and the following:

Grade - Grade B or C equipment as specified (see 6.1).
Operating range - The load to full load at specified input voltage range.

3.3.14 Inclined operation. - The motor generator and control equipment shall be designed to operate within specification limits under the following conditions:

(a) Type II motor generators:
Motor generator shaft inclined 45 degrees, generator end low for 1 hour.
Motor generator shaft inclined 45 degrees, motor end low for 1 hour.
Control equipment inclined 45 degrees in any direction for 1 hour.

(b) All other motor generators:
Motor generator shaft inclined 15 degrees, generator end low for 0.5 hours.
Motor generator shaft inclined 15 degrees, motor end low for 0.5 hours.
Control equipment inclined 15 degrees in any direction for 0.5 hours.

3.4 Motor generator. -

3.4.1 General. - Each motor generator shall be of the two bearing type with the stationary component parts of the motor and the generator, rigidly coupled together in such a fashion that the combined unit will

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Table IV - Minimum material requirements.

Item	Material	Specification	Limitations
Brush,	Various	MIL-B-3743	See MIL-B-3743
Caps and cartridge	Steel	See Note 1	
Ball bearing end brackets, terminal boxes, covers, frames flanges, vee rings, shells, drip covers baffles, hand hole as access covers	Malleable iron	QQ-I-666	
	Nodular Graphitic iron	MIL-I-17166	
Eye bolts	Steel		
Fans	Steel	See Note 1	
	Aluminum		
Grease	Petroleum	MIL-G-3545	Bearings operating at 110°C and below
	Silicone	MIL-L-15719	Bearings operating at total temperature above 110°C
Holders, brush bodies, frames, fingers	Brass		
	Bronze		
	Steel		Corrosion resistant treated for corrosion resistance
Springs	Steel, carbon		Cadmium plated and baked after forming
	Steel corrosion resistant		
	Copper-berryllium		
Shafts	Steel		Tensile strength not less than 75,000 p.s.i. and elongation not less than 20 percent in 2 inches
Spider and quills	Steel	See Note 1	
Studs, brush holder	Steel	See Note 1	Treated for corrosion resistance
	Brass	Commerical	

Note -1.

Unless otherwise indicated in table IV, steel parts may be cast, fabricated, wrought or forged. Cast steel shall be in accordance with Class B of Specification MIL-S-15083, or equivalent, except that radiographic and magnetic particle inspection will not be required.

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be capable of maintaining proper alignment without dependence upon the supporting structure of the ship, and shall not require any additional structural bedplate. The rotating elements of the motor and generator shall be mounted on a single one piece shaft. To remove or insert the rotor it shall be unnecessary to remove any stationary windings or remove any motor generator mounting bolts.

3.4.2 Mounting.

3.4.2.1 Mounting for type I motor generators. - Type I motor generator shall be designed for horizontal deck mounting. The number, size, and spacing of mounting bolts shall be as indicated in Appendix III.

3.4.2.2 Mounting of type II motor generators. - Type II motor generators shall be designed for horizontal deck or overhead mounting. In addition, they shall be so designed that they can be mounted on four resilient mounts in accordance with Specification MIL-M-17508, MIL-M-17191 or MIL-M-19379 as applicable. The mounting arrangement shall be such that the mounting pad of the ship's foundation is on a plane passing through the center of gravity of the motor generator. Drip shields, where provided, shall be removable and reversible for mounting with either the overhead or the deck mounting.

3.4.3 Dimension and weight. - The motor generators shall be as light as possible consistent with specified performance and design. The dimensions and weight of type I motor generators shall be as specified in Appendix III. The dimensions and weight of type II motor generators shall be as specified (see 6.1).

3.4.4 Internally excited vibration. -

3.4.4.1 Mechanical balance. - Motor generators shall be balanced at any load and speed in the operating range. Windings not in mechanical symmetry shall have dummy coils. In general, the proper mechanical balance shall be effected by the use of balance weights securely attached, (if bolts are used, corrosion-resistant bolts and locknuts shall be used), removal of material, securely welded steel weights, or by metal carried in a receiver in such a manner as to preclude its breaking loose. Balancing by the use of solder on the banding wire is permissible subject to the following limitations:

(a) Diameter of the rotor not to exceed 16 inches.
(b) Peripheral speed not to exceed 6,500 feet per minute at rated load.
(c) All armature cores to be balanced before the windings are inserted.
(d) Only tinned banding wire shall be used.
(e) The solder shall not cover an arc of more than 25 percent of the circumference of the armature measured on the banding wire unless satisfactory to the bureau or agency concerned. Where the manufacturer prebalances the armature and takes care in placing the windings in the slots to reduce unbalance due to axial misalignment of the coils, this value may be increased to 33 percent if satisfactory to the bureau or agency concerned.
(f) Thickness of solder over banding wire not to exceed 3/32 inch, and in no case to extend past the center of the air gap.
(g) When metal strips or wedges are used in the rotor slots for final balancing they shall be well insulated from the rotor core.

3.4.4.2 Balance limits for type I motor generator. - The maximum allowable amplitude in inches for type I motor generators shall be as specified in table V.

Table V - Mechanical balance.

Weight of motor generators	Maximum allowable peak to peak amplitude
	Balance
Pounds	inch
Up to 350	0.0005
From 351 to 800	.00075
From 801 to 2,000	.001
From 2,001 and up	.0015

3.4.5 Speed. - The speed of the motor generator shall be at the manufacturer's discretion except that DN values (bearing bore in MM X r.p.m.) exceeding 200,000 shall not be used, and those rated at 60 kw. and larger shall be designed for operation at speeds not greater than 1800 r.p.m. in the case of 400 cycle generators and 2,000 r.p.m. in the case of 400 cycle generators. The design of the motor generators shall be such that they can safely withstand an overspeed of 125 percent of the rated speed.

3.4.6 Inherent speed regulation. - The design of the motor generator shall be such that while operating in manual control with any fixed voltage in the input voltage range specified in 3.2.2 there will be no rise in speed with incremental increases in load and the system shall be stable.

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3.4.7 Bearings.-

3.4.7.1 Type.- Medium series ball bearings of the following types shall be used:

(a) For type I motor generators, ball bearings shall be type 120, medium series in accordance with Specification FF-B-171. A plate bearing the warning "DO NOT LUBRICATE" shall be secured to the end brackets.
(b) For type II generators, ball bearings shall be in accordance with Specification MIL-B-17931 and shall be medium series. If type 120 bearings are used a warning plate as specified in 3.4.7.1(a) shall be provided.

3.4.7.2 Mounting diameters and tolerances.- The shaft, shaft shoulder, housing and housing shoulder diameters shall be as follows:

(a) For type I motor generators - as specified in table VI.
(b) For type II motor generators - as approved by the bureau or agency concerned.

Table VI - Bearing shaft, shaft shoulder, housing and housing shoulder diameters

Bearing	Shaft diameter inches		Shaft shoulder diameter inches	Housing diameter inches		Housing shoulder diameter inches
	Minimum	Maximum	Minimum	Minimum	Maximum	Maximum
305	0.9844	0.9847	1.140	2.4409	2.4416	2.280
306	1.1812	1.1815	1.340	2.8346	2.8358	2.620
307	1.3781	1.3785	1.590	3.1496	3.1503	2.940
308	1.5749	1.5753	1.813	3.5433	3.5442	3.310
309	1.7718	1.7722	2.000	3.9370	3.9379	3.660
310	1.9686	1.9690	2.270	4.3307	4.3316	4.000
311	2.1655	2.1660	2.468	4.7244	4.7253	4.380
312	2.3623	2.3628	2.688	5.1181	5.1191	4.750
313	2.5592	2.5597	2.968	5.5118	5.5131	5.190
314	2.7560	2.7565	3.156	5.9055	5.9068	5.560
315	2.9529	2.9534	3.380	6.2992	6.3005	5.940
316	3.1497	3.1502	3.550	6.6929	6.6942	6.250
317	3.3466	3.3472	3.840	7.0866	7.0879	6.620
318	3.5434	3.5440	4.060	7.4803	7.4817	7.000
319	3.7403	3.7409	4.290	7.8740	7.8754	7.380
320	3.9371	3.9377	4.488	8.4646	8.4660	7.940

3.4.7.3 Bearing pre-load washers or springs.- The use of bearing pre-load washers and springs or both is preferred provided the washer and spring, or both are secured in the housing to preclude its falling out of the end bracket during repairs and maintenance of the motor generator.

3.4.7.4 Mounting.- The bearings shall be secured on the shaft by means of locknuts and lockwashers. One end bearing shall be fixed axially in the housing by suitable housing and end cap shoulders. The axial movements of the shaft shall be not more than 0.010 inch including bearing end play for bearings up to and including size 310 and .015 inch for bearings over size 310.

3.4.7.5 Heat stablization.- Bearings operating above 110°C shall be heat stabilized so that their dimensions do not increase more than 0.0001 inch per inch of diameter during 2500 hours at 149°C.

3.4.7.6 Housing construction.- The bearing housing shall be constructed to permit ready removal of the end bracket without the necessity of removing the bearings from shaft. The housing shall have sufficient metal to permit redrilling and bushing.

3.4.7.7 Removal of bearing.- The design shall be such as to permit removal of the ball bearing by pulling on the inner ring of the bearing with one of the tools contained in a complete Naval shipboard set of tools as specified in Specification GGG-P-781.

3.4.8 Fans.- Fans shall be keyed to the shaft to prevent rotation and shall be secured in a manner to prevent axial displacement of the fan along the shaft. Shoulders, lockwashers and nut, sleeves or clamping bolts (in the case of a split fan) may be used. Shrink rings or press rings may be used where the fans do not have to be removed from the shaft during disassembly of the motor generators.

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Locked set screws may be used only where closed and key seats or woodruff keys are used.

3.4.9 Cores.- Cores shall be secured to the shaft, spider or frame to prevent their rotational and axial displacment under operational or test conditions. A pin through the shaft is not an acceptable means of preventing axial displacement. Welding shall not be used to secure the core or quill to the shaft. The use of spiders constructed of individual flutes to the shaft is acceptable when preheat and stress relieving procedures satisfactory to the bureau of agency concerned are employed. Press or shrink fits without keys shall be approved by the bureau or agency concerned.

3.4.10 Commutators.-

3.4.10.1 Securing.- The commutator shall be secured rigidly to the shaft or spider in a manner as to positively prevent rotational or axial movement of the commutator with respect to its support. A key shall be used to prevent rotational movement. Axial movement shall be prevented by the use of a nut with a locking device, interference fit or a locking ring. When an interference fit is used to prevent axial movement, the minimum interference shall be as specified in table VII.

Table VII - Minimum interference fits for ring assembly shell to shaft, and commutator assembly to shaft or spider.

Shaft diameter at assembly section	Minimum interference fits
Inches	Inch
Over 0 to 2, inclusive	0.0003
Over 2 to 3, inclusive	.0006
Over 3 to 4, inclusive	.0009
Over 4 to 5, inclusive	.0013
Over 5 to 6, inclusive	.0015
Over 6 to 7, inclusive	.0019
Over 7 to 8, inclusive	.0024
Over 8 to 9, inclusive	.0028
Over 9 to 10, inclusive	.0037
Over 10 to 11, inclusive	.0042
Over 11 to 12, inclusive	.0046
Over 12 to 13, inclusive	.0051
Over 13 to 14, inclusive	.0056
Over 14 to 15, inclusive	.0061
Over 15 to 16, inclusive	.0065
Over 16 to 17, inclusive	.0070

3.4.10.2 Copper segments.- The copper segments shall be securely retained in such manner as will prevent their relative displacement as a result of centrifugal forces and the stresses imposed by repeated expansion and contraction in service. The segments, when worn to the full extent of the allowable wearing depth specified in table VIII, shall be of ample section to operate satisfactorily.

Table VIII - Wearing depth of commutators and collector rings

Commutator diameter (inches)	Permissible reduction in diameter
	Inch
2 and under	3/16
Over 2 and up to 4	3/8
Over 4 and up to 6	1/2
Over 6 and up to 8	5/8
Over 8 and up to 12	3/4
Over 12 and up to 15	7/8
Over 15	1

3.4.10.3 Sealing.- After the assembly of the commutator all crevices or joints at each end between the bars and the retaining flanges shall be completely filled up and sealed over with high-grade, flexible, baking insulation varnish in such a manner as to prevent the entrance of moisture oil, carbon, or copper dust, or other deleterious substances at these points; likewise, the creepage path from the ends of the copper bars to the metal flanges shall be given a heavy, durable coating of flexible, baking insulating varnish as a protection against short circuits and ground at these points.

3.4.10.4 Mica undercutting.- The mica insulation between bars shall be undercut on all motors. The undercutting shall be accomplished by removing all mica between bars to form a groove not to exceed 3/64 inch in depth.

3.4.10.5 Curing.- All commutators shall be cured at higher than operating temperatures and shall be of such thoroughly solid construction as will insure that they will hold their shape in service and obviate the necessity of frequency dressing the commutators on account of high bars or flat spots.

3.4.10.6 Tightening.- Suitable provision shall be made for tightening the commutator retaining flanges without the necessity of removing the commutator from the shaft or spider.

3.4.11 Collector rings.- Collector rings shall run true, have a smooth polished surface, and be free from porosity or hard spots. The rings with their insulation shall be shrunk over a steel sleeve, and shall be of such thoroughly solid construction throughout as to insure against their becoming loosened or eccentric due to changes in temperature

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in service. The collector rings shall be secured rigidly to the shaft or spider in a manner as to positively prevent rotational or axial movement with respect to the shaft. Keys shall be used to prevent rotational movement. Axial movement shall be prevented by the use of a nut with a locking device, an interference fit or a locking ring. When an interference fit is used to prevent axial movement, the minimum interference shall be as specified in table VII. The wearing depth of collector rings shall be as shown in table VII.

3.4.12 Brush rigging. -

3.4.12.1 Accessibility. - Brush holders shall be readily accessible for adjustment, replacement or renewal of the brushes and springs. It shall not be necessary to remove the end brackets to replace or adjust the brush holders or brushes.

3.4.12.2 Adjustability of commutator brush holder rigging. - The rigging shall be of such design that the yoke or rocker ring is adjustable around the periphery of the commutator. The setting of the rigging shall be marked permanently on the yoke or rocker ring and the frame or end bracket.

3.4.12.3 Mounting of brush holders rigging. - When brush holders are secured to the yoke or rocker ring the yoke or rocker ring shall be secured by one of the following methods:

(a) A split type hub clamped about a machined hub in the end bracket on bearing housing.
(b) Bolts passing through slotted holes in the yoke or rocker ring. If this method is used, the axial movement of the yoke or rocker ring shall be restrained by means of machined fits between the yoke or rocker ring and the end bracket or bearing housing. The bolts shall only be used to lock or ring in the proper rotational position.

3.4.12.4 Staggering. - Commutator brush holders shall be staggered in pairs so as to insure even wear of the commutator.

3.4.12.5 Mounting of collector ring brush holders. - Collector ring brush holders maybe mounted on insulated studs attached to the end brackets.

3.4.12.6 Alignment. - Brush holder shall be so mounted and aligned that brushes are parallel to the shaft and do not extend beyond the edge of collector ring or commutator.

3.4.13 Brush holders. -

3.4.13.1 General. - Contact surfaces shall not react chemically to produce corrosion and sticking of brushes when brushes or graphite, electro graphite, electro graphite or copper-graphite are used. The brush holders shall be of such design that the brushes will not bind in the brush holders. The brush-feeding device shall ensure proper brush contact under all normal conditions of brush and commutator or collector ring wear.

3.4.13.2 Brush stop. - Means shall be provided to prevent brushes from being thrown out of their brush holders as a result of severe shock or from being displaced to such a position that they may bind in their brush holders and not return to their normal operating position. The devices used shall not unduly complicate replacement of the brushes, require the use of a special design of brushes, or result in cumbersome or intricate design of brush holders.

3.4.13.3 Pressure finger. - The pressure finger shall be of such design that when operating with a brush having a smooth top surface (not notched) or a hammer clip, it will not slide off the brush or clip under any operating conditions. When the brush is omitted from the holder, a stop shall prevent the finger from making contact with the commutator or collector ring.

3.4.13.4 Brush holder spring. - Springs shall be of either the adjustable type or constant pressure type. Where adjustable springs are used means shall be provided for adjusting the brush pressure without special tools.

3.4.13.5 Types. - Brush holders shall be either full box type or open front type. The open front type shall be of such design that the frame work covers not less than 50 percent of the front side.

3.4.13.6 Terminal screw. - A separate terminal screw shall be provided for securing the brush shunt terminal lugs to the brush holder.

3.4.14 End brackets. - An accurately machined shoulder joint shall be provided between the frame and the end brackets. End brackets shall be secured to the frame by not less than four hexagon-head bolts. Resilient gaskets shall not be placed between any bearing support member and the frame. Openings of adequate size and number shall be provided so as to give easy access to and a direct view of commutator, collector rings and brushes while the machine is in operation. These openings shall be provided with substantial translucent covers. The enclosing covers (hand-hole or access-hole covers) shall be

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readily removable. The engaging bolts or locking devices shall be attached to the frame, the end bracket, or the cover as the design permits, in such a manner as to prevent the bolts or locking devices from being misplaced or dropped into the machine during repair operations. Ready access shall be provided to permit measurement of the motor and generator air-gaps as specified in 4.6.4 and for the measurement of shaft speed.

3.4.15 Feet. - Feet or mounting pads shall be machined on the mounting surfaces and shall be either cast integral with or welded to the frame, except that mounting pads for type II motor generators may be detachable and secured with bolts if doing so will permit the motor generator to be passed through a 25 inch diameter submarine hatch. The feet or pads shall be provided with round holes for holding-down bolts.

3.4.16 Air gap. - Adjustable air-gap tolerances shall not exceed 10 percent of the nominal air-gap for each particular group of poles.

3.4.17 Field poles - Except for 400 cycle machines, field poles shall be so attached as to permit ready removal of individual poles. Field poles for 400 cycle machines may be an integral part of the complete rotor or stator lamination and need not be individually removable. Stationary field poles shall be capable of being removed from the frame without removing the rotating assembly, and shall be attached to the frame so that the air gap under each pole is adjustable. The method of securing rotating field poles to the shaft or spider shall result in a factor of safety of not less than 2.0, based in the elastic limit of the material and the rated speed of the motor generator. Commutating poles shall be furnished for all d.c. motors.

3.4.18 Field coils. - Field coils shall be thoroughly insulated from adjacent conducting and grounded parts. All coils shall be completely 1/2 lap wrapped with tape before the varnish dip and bake. Field coils shall be form wound. This shall not be interpreted as precluding the use of the pole piece as the form. All field coils in the same machine shall be readily interchangeable. The coils shall be so secured that they cannot become loosened or damaged by vibration, or produce vibration by a slight shifting in coils position.

3.4.19 Coil connections. - All coils connections shall be made on the commutator end of the motor or the collector ring end of the generator. Connections between winding end to commutator shall be effectively soldered, or brazed and, if separate risers are used, they shall be mechanically and electrically connected to the commutator bar in a manner satisfactory to the bureau or agency concerned. All motor field coil connections shall be bolted connections.

3.4.20 Excitation requirements. - The motor and generator design, construction, assembly, and quality control exercised shall be such that the ampere turn requirements for the motor and generator at any load do not differ more than plus or minus 10 percent from the design or nominal value.

3.4.21 Stray field reduction. - To reduce the stray magnetic field the motor generator shall incorporate the following:

(a) Stray magnetic fields caused by electrical circuits shall be kept to a minimum by making the area of electrical current loops as small as possible and arranging adjacent loops so that their fields will be opposition. Current loops shall enclose no magnetic material unless essential for proper performance of equipment. All motor and generator windings and connections shall be balanced for ampere-turns around the shaft to eliminate stray fields insofar as possible. Each circuit which makes even part of a turn around the machine shall return on itself back to the starting point, with conductors for the forward and reverse currents arranged at the same end of the machine and as close as adequate insulation permits. All connections shall be as short as possible.
(b) Brush rings shall be complete rings of uniform resistance throughout.
(c) Leads to the positive and negative brush rings shall be located as close together as possible.

3.4.22 Radio interference reduction. - To reduce the radio interference the motor generator shall incorporate the following:

(a) Commutating pole windings of balanced magnetic design.
(b) Compensating windings.
(c) Largest practical number of armature coils and commutator bars.
(d) Shallow armature slots.
(e) Short-pitch armature windings.
(f) Placing half of the motor series windings on each side of the commutator to act as a choke.
(g) Minimum eccentricity of commutators and collector rings.
(h) Balanced design to eliminate shaft and bearing currents without the use of auxiliary brushes.

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3.4.23 Terminal boxes.- Terminal boxes shall be provided and shall be securely bolted or welded to the frame. For type II motor generators the terminal boxes shall be of the removable type and the design shall be such that they can be mounted on either side of the motor generator. The boxes shall not have any openings but be suitable for drilling and installing cable clamps or stuffing tubes by the installing activity. Terminal boxes not readily removable from the motor generator shall be provided with removable plates for the purpose of drilling as specified herein. Where the controls are mounted on top of the motor generator, separate terminal boxes shall not be furnished, but all connections shall be made in the control cabinet on the top of the motor generator. Removable cable entrance plates shall be provided on both sides of the control cabinet. Gaskets shall be furnished under all terminal box covers; removable plates; and between the box and motor generator frame, if the box is bolted to the frame.

3.4.24 Terminal boards.- Terminal boards are not required on motor generators up to and including 10 kw. which do not have controls mounted atop the motor generator. Motor generator leads shall terminate within the terminal box with at least 12 inches of lead cable provided within the terminal box. On motor generators over 10 kw. and where the controls are mounted atop the motor generator, or both, the leads shall terminate at a terminal board within the terminal box. The terminal board shall be securely mounted within the terminal box in such a manner as to permit connection of the internal leads to the back of the board and the external leads to the front of the board. Terminal studs shall be secured to the boards in a manner to prevent turning of the studs when tightening or loosening the nuts used to secure lead connectors. Terminal identification corresponding to the lead identifications specified in Standard MIL-STD-195 shall be permanently marked on the front of the terminal board.

3.4.25 Terminal leads.- All terminal leads shall be of flexible, stranded cable or wire. An insulating sleeve or other suitable means shall be provided to prevent abrasion of the lead insulation by metallic edges of the motor generator frame, terminal box, piping or cable clamps.

3.4.26 Securing terminal leads.- Leads shall be secured in a reliable manner to prevent their coming in contact with moving parts or being chafed by contact with stationary parts. Leads shall not be secured to end bells or in any other manner which would prohibit ready removal of the end bells or rotating assembly. Terminal leads shall be secured to the terminals on the terminal board, brush holders, and collector rings by means of solderless type connectors. Jam nuts or lockwashers and nuts shall be provided to prevent the connectors from loosening under shock or vibration. Securing of leads to brush holders by means of special provisions in the design of the brush holder will be considered as an exception subject to approval of the bureau or agency concerned.

3.4.27 Lead identification.- All motor and generator leads shall be permanently marked in accordance with Standard MIL-STD-195. In order to achieve standard marking of all leads, all motor generators shall rotate clockwise when facing the generator end. The direction of rotation shall be permanently marked on the frame of the motor generator by means of an instruction plate stating "rotation" with an arrow in the direction of rotation.

3.4.28 Identification plate markings.- A separate plate for each motor and generator or a single plate applicable to both the motor and generator may be furnished. The minimum data to be marked on the identification plate (or plates) shall include the following:

(a) Major unit name (that is d.c. motor, a.c. generator or d.c./a.c. motor generator as applicable).
(b) Federal Stock number.
(c) Manufacturer's name and serial number.
(d) Technical manual number (NAVSHIPS _____).
(e) Contract number and year of manufacture.
(f) Blank space for Government inspector's official acceptance stamp.
(g) Salient design characteristics of motor, namely: horsepower, voltage (nominal voltage and voltage range shall be given for type II) current type of winding, speed, class of insulation, maximum ambient temperature (nominal current and current range shall be given for type II).
(h) Salient design characteristics of generator, namely, kw., power factor, voltage, current, frequency, number of phases, class of insulation and maximum ambient temperature.

3.4.29 Permanent marking.

3.4.29.1 Frame.- The manufacturer's name, serial number and the Federal stock number (where available) shall be stamped in the frame under the identification plate.

3.4.29.2 Rotating element.- The rotating element shall be marked so as to identify the manufacturer's serial number and the Federal stock number (where available) shall be stamped on the shaft.

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3.5 Control equipment. -

3.5.1 General. - Control equipment requirements in addition to those which affect the combined system performance specified in 3.2 and the detailed requirements in 3.3 shall be as specified herein.

3.5.2 Arrangements. - Control cabinet shall be furnished in either one of the following arrangements as specified. (see 6.1).

(a) Arrangement A-
A complete control panel in which all components are mounted within a single enclosure.

(b) Arrangement B-
Separately mounted components in which the major components are mounted separately within their own enclosures and the smaller components are mounted on the ship's main or auxiliary switchboard.

(c) Arrangement C-
Consolidated mounting of components in which two or more of the major components are mounted within one enclosure and attached to the frame of the motor generator with any remaining major components mounted separately within their own enclosures. The smaller components are mounted on the ship's main or auxiliary switchboard.

3.5.2.1 Control panel mounted control equipment. - When control Arrangement A is specified the following equipment and any other equipment necessary for satisfactory control of the motor generator shall be mounted in a single control panel:

(a) Motor controller.
(b) Voltage regulator and static exciter.
(c) Frequency regulator.
(d) Circuit breaker.
(e) Manual frequency control rheostat.
(f) Manual voltage control rheostat.
(g) Voltage and frequency automatic - manual transfer switches.
(h) Voltmeter for a. c. generator output.
(i) Ammeter for a. c. generator output.
(j) Frequency meter for a. c. generator output.
(k) Regulated voltage adjusting rheostat.
(l) Regulated frequency adjusting rheostat.
(m) Synchroscope.
(n) Motor controller "start-stop" push button.
(o) Motor controller, overload relay "reset" push button.
(p) D. c. ammeter to measure d. c. input (type II).

3.5.2.2 Separately mounted control equipment. When control Arrangement B is specified the following equipment and any other equipment necessary for satisfactory control of the motor generator:

(a) Motor - controller.
(b) Frequency regulator.
(c) Voltage regulator and static exciter.
(d) Manual frequency adjusting rheostat.
(e) Manual voltage adjusting rheostat.
(f) Regulated frequency adjusting rheostat.
(g) Regulated voltage adjusting rheostat.
(h) Voltage and frequency "automatic-manual" transfer-switch.
(i) Motor controller "start-stop" push button.

3.5.2.3 Consolidated mounting of equipment. - When control Arrangement C is specified the equipment specified in 3.5.2.2 shall be furnished except that components of the frequency regulator, voltage regulator and static exciter are mounted on top of and in the airstream of the motor generator.

3.5.3 Enclosures. -

3.5.3.1 Dimensions and weights. - The control equipment enclosures shall be as light and compact as possible, consistent with specified performance and design. For control, Arrangement A the weights and dimensions shall be as specified in Appendix III. For control Arrangements B and C, the weights and dimensions shall be as specified, (see 6.1). All controls for type II motor generators shall be capable of passing through a 25 inch diameter submarine hatch. However, if it is not practical to provide control cabinets which will pass through a 25 inch submarine hatch the components of the control equipment shall be mounted on panels which will pass through the hatch.

3.5.3.2 Mounting. - All control equipment enclosures shall be suitable for either bulkhead or deck mounting as specified (see 6.1). Mounting holes shall be round and have a diametrical clearance of 1/16 inch. Mounting bolts shall not be less than 5/8 inch in diameter. The enclosure shall not be supported from more than one plane.

3.5.3.3 Enclosure face. - When control Arrangement A is furnished; all instruments, switches, rheostats, circuit breakers, and other components needed for normal control of the equipment shall be mounted on the front of the panel. The automatic voltage and frequency level adjusting controls are not considered normal adjustments and shall not be mounted on the front of the control panel, but at a convenient location within the panel. The meters shall be mounted at the top of the panel.

3.5.3.4 Accessibility. - All parts which require servicing, repair, replacement, or periodic

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adjustment during the life of the equipment shall be readily accessible through the front of the panel or enclosure. Accessibility may be obtained by providing hinged doors or panels, swing-out or intermediate panels, or drawer type construction. Wherever practicable, subassemblies shall be utilized in the design of the equipment in order to reduce the time required for service and maintenance, for example, separate assemblies for the voltage detector, frequency detector, individual amplification stages. The use of plug in assemblies or quick-disconnect devices shall not be incorporated in the design. Bulkhead mounted control enclosures for type II motor generators shall be provided with hinged doors only, control enclosures mounted atop the motor generator shall be provided with either removable panels or hinged doors as approved by the bureau of agency concerned.

3.5.3.5 Positioning devices.- Each access panel, cover or door not designed for complete removal (for example, hinged or sliding doors) shall be provided with stops and positioning devices to automatically stop it in its open positions and to prevent its moving from those positions due to gravity or ship motion. Its full open position shall be at least enough to permit unimpeded access and shall minimize its obstruction of working or passage areas. Intermediate stops shall be provided when added convenience for maintenance will result.

3.5.3.6 Internal temperature.- The design of the equipment enclosure and the placement and selection of parts shall be such that no part will be exposed to a higher operating temperature than the maximum rated temperature for the part.

3.5.3.7 Ventilation.- Forced draft ventilation shall not be used except where the components are mounted in the intake airstream of the motor generator (resistors may be mounted in the motor generator exhaust airstream).

3.5.3.8 Heating.- Circuit design shall not be such as to require the use of heaters to obtain proper operation under conditions of varying ambient temperature or the variation in temperature rise within the enclosure due to heating of components under different operating conditions.

3.5.3.9 Guard rails.- All deck mounted control panels shall be provided with horizontal guard rails or vertical grab rods located so that an operator will have means of support while operating the equipment panel. The rail or rod shall be of smooth hardwood, or other non-conducting material, supported with a sufficient number of brackets to provide adequate support for personnel under all operating conditions of the vessel. The rail shall be attached in such a manner that all panels can be removed, and hinged panels opened, without removal of the rail. These rails shall be located so that they will not interfere with operation of the equipment and shall not extend more than 6 inches from the front of the panel. The rail shall be capable of resisting a force of 400 pounds applied at any point on the rail without permanent distortion or damage to the rail or supporting means.

3.5.3.10 Operating and maintenance instructions.- A reduced size copy of abbreviated operating instructions, complete schematic diagram of the system and wiring diagram shall be laminated in plastic and stowed in the control equipment panels or enclosures. The diagram printing process shall be such that fading of the recorded information will not occur. The diagrams and instruction sheets shall be either mechanically attached to enclosure doors or stowed in racks in the enclosure in such a manner as not to interfere with operation of the equipment.

3.5.4 Component requirements.-

3.5.4.1 Component mounting.- In order to prevent the possibility of displacement of components due to shock and vibration, the use of slotted mounting bolt holes is prohibited. The holes shall be round with a minimum diametrical clearance over the mounting bolt diameter. In no case shall the diametrical clearance be greater than 1/16 inch. Components previously approved by the bureau or agency concerned shall be mounted in a manner similar to the way they were mounted when subjected to the HI shock and vibration tests during their qualification testing.

3.5.4.2 Component and terminal markings.- All components and terminals shall be marked with identifying designations shown on master drawings approved by the bureau or agency concerned. Wherever practicable, the designations shall be in accordance with Standard MIL-STD-195. The identifying designations may be made by metal stamping, engraving, silk screening, stenciling, or rubber stamping; however, silk screening, stenciling, or rubber stamping may not be used on terminal boards.

3.5.4.3 Adjustable circuit elements.- The number of adjustable circuits shall be kept to an absolute minimum and limited to those adjustments required by maintenance personnel. When 100 percent adjustable resistors are used, a fixed series resistor shall be employed to protect the circuit components from overload due to improper adjustment of the variable resistor. A complete description and instructions for such adjustment shall be included in technical manuals.

3.5.4.4 Mechanical stability.- All controls of the equipment shall be so designed that when the

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equipment is properly installed and whether operating or nonoperating, the setting, position, or adjustment of any control shall not change due to normal service conditions.

3.5.4.5 Adjustment locking devices. - Adjustment locking devices, where used, shall be capable of retaining the controls in any given setting within the range of control under shock and vibration. The locking and unlocking action shall be easily and quickly accomplished with the equipment energized and shall not affect the setting of the control. When in the unlocked position the locking device shall not interfere with the normal operation of the control. Where vernier controls are used, the locking device shall operate on both main and vernier controls.

3.5.4.6 End stops. - Each control which is not capable of an unlimited degree of motion without damage shall be provided with positive end stops to limit the degree of motion of the control to that within its normal excursion or within its range of calibration. Where flexible control shafts are employed, end stops shall be provided on the driving end of the shaft.

3.5.4.7 Motor controllers. - Except as otherwise specified herein, the d.c. motor controller and master switches shall be in accordance with Specification MIL-C-2174 and the following:

(a) Voltage - The nominal voltage shall be as specified in 3.2.2.
(b) Duty - continuous.
(c) Operation - magnetic.
(d) Type - resistor.
(e) Performance - semiautomatic.
(f) Size - standard sizes as required by total input requirement for the power supply.
(g) Protection features - undervoltage protection (LVP).
(h) Push button location - see 3.5.2.1 and 3.5.2.2.
(i) Ammeter shunt - a 50 mv shunt shall be provided in type II motor controllers for motor generators when control arrangements B or C are specified.

3.5.4.8 Voltage regulator. - The voltage regulator and excitation system shall be designed to meet the system performance requirements specified in 3.2 when operating with its intended motor generator and as follows:

(a) Type - static. That is, one which does not employ vibrating contacts, carbon piles, electron tubes or mechanically-moving parts such as relays and tachometers. This restriction does not apply to switching devices required to start the motor generator or to synchronize and inter-connect the motor generators for parallel operation. Sliding contact variable resistors and rheostats other than those varied only by manual means will not be permitted. The required performance shall be obtained through circuits using transformers, reactors, magnetic amplifiers, transistors, rectifiers, resistors and capacitors.
(b) Circuit controller. - The specified performance requirements shall be met by controlling the generator main fields.
(c) Source of excitation power. - The generator excitation power shall be furnished from rectified generator output. Rotating exciters shall not be used.
(d) Field flashing. - Field flashing shall be accomplished from the d.c. supply to the motor generator in accordance with one of the following:

(1) Provide special static circuits which will provide sufficient excitation for the generator voltage to build up when the motor generator is started while in "automatic" control.
(2) Operate the field flashing circuit from the motor generator starter. Where the power for flashing the generator field is obtained from a source whose voltage is in excess of that required to cause a build up of the generator voltage, reduction of the voltage to this value for field flashing shall be obtained from a voltage dividing network so that at the instant the field is flashed, full source voltage is not applied across the slip ring film.

(e) Compensation for parallel operation. - The reactive load division circuits provided to meet the parallel operation requirements specified in 3.2.28.1 shall be automatically energized through auxiliary contacts on the generator circuit breaker so that generators being brought on will automatically be paralled with the generator (or generators) already on the line when its circuit breaker is closed. The reactive load division circuits shall be adjustable to compensate for differences in reactive load sharing characteristics of the generators. The adjusting means shall be located on the inside of the voltage regulator enclosure.

3.5.4.9 Frequency regulator. - The frequency regulator and excitation system shall be designed to meet the system performance requirements specified in 3.2 when operating with its intended motor generator and as follows:

(a) Type - Static type generator frequency sensing (See 3.5.4.8a.)

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(a) Type - Static type generator frequency sensing (See 3.5.4.8a.)
(b) Circuit controlled - The specified performance requirements shall be met by controlling the motor main or auxiliary fields or both.
(c) Source of excitation power - The motor excitation power shall be furnished partially from the generator output and partially from the d.c. input power supply.
(d) Overspeed protection - The frequency regulator shall be designed in such a manner that any failure of the frequency regulator shall not cause overspeeding of the motor. Speeds in excess of 115 percent rated shall be considered overspeeding. Type II motor generators shall be provided with an overspeed trip device to shut down the motor generator in the event the speed should exceed 115 percent rated.
(e) Compensation for parallel operation- Cross current compensation shall be provided to meet parallel operation requirements for real load division specified in 3.2.28.1. The method of energizing these circuits shall be compatible with that specified in 3.5.4.8(e).

3.5.4.10 Manual voltage control. - In order to keep the size of the manual voltage control rheostat to a minimum, the excitation of the generator under manual operation shall be provided through the last stage of magnetic amplifier of the voltage regulator. Only minimum number of voltage regulator circuit shall be energized during manual operation.

3.5.4.11 Manual frequency control. - Manual frequency control shall be provided by controlling the excitation of the motor through the last stage of the magnetic amplifier of the frequency regulator.

3.5.4.12 Rheostats and resistors. - Resistors and rheostats rated 50 watts or less shall be in accordance with Specification MIL-R-15109 or MIL-R-22. Rheostats rated above 50 watts shall be in accordance with Specification MIL-R-15109.

3.5.4.13 Capacitors. - Except as noted herein, all capacitors shall be paper-dielectric fixed capacitors of the extended-foil construction conforming to the requirements of Specification MIL-C-25. Styles and ratings listed in Standard MIL-STD-242 shall be used where possible. For d.c. applications, the sum of the highest d.c. voltage level plus the peak a.c. amplitude voltage to be applied shall not exceed 70 percent of the working voltage rating of the capacitor. This 70 percent rerating factor shall be in addition to voltage derating with temperature as required by individual specification sheets under Specification MIL-C-25. The above 70 percent rerating factor for d.c. applications shall be reduced to 60 percent for 60 cycle a.c. applications and to 50 percent for 400 cycle applications. For high values of capacitance (above 200 microfarads) where severe space and weight restrictions are imposed dry-electrolytic aluminum foil fixed capacitors conforming to the requirements of Specification MIL-C-62 are permitted with the following limitations:

(a) Use only in d.c. circuits.
(b) Do not derate in voltage (from ratings available, select one equal or as nearly so as possible to the maximum voltage to be imposed on the capacitor by the conditions of the application).
(c) Select capacitors of the plug-in-type and mount them with a retaining device. These shall be of the styles and ratings listed in Standard MIL-STD-242 where possible.
(d) Do not apply if ambient temperature at the capacitor can be expected to go above +85°C or below - 40°C.

3.5.4.14 Rectifiers. - Rectifiers shall be either silicon or selenium types. When both types are suited to an application, the order of preference shall be silicon over selenium.

3.5.4.14.1 Silicon rectifiers. - Individual silicon rectifiers shall be selected from Appendix II. In addition, they shall comply with the requirements of the individual specification sheets and maximum rating and rerating values specified in table I of Appendix II. Non-listed silicon rectifiers will require specific approval as required by Specification MIL-E-317 for non preferred parts. In addition, they shall either be covered by an individual specification sheet or they shall comply with the requirements of an individual specification sheet listed in Appendix II for a similarly constructed device (axial lead, stud-mounted, glass package, and so forth) except that the device manufacturer's latest published commercial specification values applicable to the device (or approved values based on the commercial product) shall be substituted for the characteristic value contained in the applicable specification sheet. Individual diodes shall be accessible for replacement.

3.5.4.14.2 Selenium rectifiers. - Selenium rectifiers shall be supplied in accordance with the requirements of Specification MIL-R-18281.

3.5.4.14.3 Silicon controlled rectifiers. - Silicon controlled rectifiers shall conform to Specifications MIL-S-19500. Specification MIL-S-19500/108 applies to nominal 16 ampere devices, Specification MIL-S-19500/168 applies to nominal 3 ampere

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devices. Other devices shall require specific approval of the bureau or agency concerned, as specified in Specification MIL-E-917, and shall comply with the requirements of the 16 ampere or 3 ampere specifications when characteristics peculiar to the devices to be used are substituted in the specification. Devices shall be individually replaceable.

3.5 4.15 Voltage reference diodes. - Voltage reference diodes shall be selected from appendix II. They shall comply with the maximum rating and rerating requirements specified therein. Nominal 10 watt, 5 to 6 volt regulator diodes having substantially zero temperature coefficient are also permitted provided they are procured to individual sheets of specification MIL-S-19500 except that the device manufacturer's latest published commercial specification values applicable to the device (or approved values based on the commercial product) shall be substituted for the characteristic values contained in that specification. All other devices will require specific approval as required by Specification MIL-E-917 for non-preferred parts; in addition, they shall either be covered by an individual specification sheet or they shall comply with the requirements of an individual specification sheet listed in appendix II for a similarly constructed device substituting characteristic values peculiar to the devices to be used. Individual diodes shall be accessible for replacement.

3.5.4.16 Voltage regulator devices. - Voltage regulator devices shall be supplied in accordance with the requirements of Specifications MIL-E-1 and MIL-S-19500 and applicable individual specification sheets thereunder as specified in appendix II. In addition to listing individual devices from which selection shall be made, appendix II also lists maximum ratings and derating factors which must be observed in application of these devices. Non listed devices will require specific approval as required by Specification MIL-E-917 for non preferred parts. In addition, they shall either be covered by an individual specification sheet, or they shall comply with the requirements of an individual specification sheet listed in appendix II for a similarly constructed device except that the device manufacturer's latest published commercial specification values applicable to the device (or approved values based on the commercial product) shall be substituted for the characteristic values contained in the applicable specification sheet.

3.5.4.17 Transistor. - Individual transistor shall be selected from appendix II. In addition, they shall comply with maximum rating and rerating requirements specified in that table. All transistors shall be procured to the requirements of Specification MIL-S-19500 and, where issued, individual military specification sheets under Specification MIL-S-19500. If not specifically covered by such a sheet, they shall comply with the tests and inspection procedures of an individual military specification sheet listed in appendix II or Standard MIL-STD-701 for a similarly constructed device except that the manufacturer's latest commercial specification values applicable to the device (or approved values based on the commercial product) shall be substituted for the characteristic values contained in the applicable specification sheet. Types not listed in appendix II will require specific approval as required by Specification MIL-E-917 for non preferred parts. The requirements stated above for preferred types shall apply for all transistors approved for use in this equipment. Individual transistors shall be accessible for replacement.

3.5.4.18 Switches. - Control switches shall be in accordance with Specification MIL-S-15291. The switching of the voltage regulator and frequency regulator from automatic to manual shall be such that the following conditions are met when in manual operation:

(a) Disconnect (or short out as necessary) all power inputs to the regulators
(b) Isolate the regulators to prevent grounding of the motor or generator bus, except that potential transformer primary windings need only be disconnected.

3.5.4.19 Circuit breaker. - Circuit breaker when required (see 3.5.2.1) shall be either type AQB and conform to Specification MIL-C-17361, or type ALB and conform to Specification MIL-C-17588 as specified (see appendix II). The instantaneous trip setting and element rating shall be as specified (see appendix III).

3.5.4.20 Voltmeter. - The voltmeters when required (see 3.5.2.1) shall be 4-1/2 inch 250 degree scale switchboard type conforming to Specification MIL-M-16034 or the 3-1/2 inch panel type conforming to Specification MIL-M-10304. For 60 cycle applications the motor shall have a full scale range of 600 volts and a red line at 440 volts or a full scale range of 150 volts with a red line scale mark at 115 volts. For 400 cycle applications the meter scale shall be the expanded type with a range of 410 to 465 volts or 110 to 120 volts. Initial accuracy shall be 0.1 percent of the midscale value.

3.5.4.21 Frequency meter. - The frequency meters when required (see 3.5.2.1) shall be 4-1/2 inch switchboard type or the 3-1/2 inch panel type. The meter shall conform to the applicable requirements of Specification MIL-M-16125. A meter scale of 59 to 61 cycles or 390 to 410 cycles shall be furnished as required. Initial accuracy shall be 0.1 percent of the midscale value.

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3.5.4.22 Ammeter. - The ammeters when required (see 3.5.2.1) shall be 4-1/2 inch 250 degree scale switchboard type conforming to Specification MIL-M-16034. The meter scale shall be as specified (see Appendix III) and have a red line scale mark at rated amperes.

3.5.4.23 Ammeter shunts. - Ammeter shunts when required (see 3.5.4.7) shall be in accordance with Specification MIL-I-1361.

3.5.4.24 Synchroscope. - Synchroscopes when required (see 3.5.2.1) shall be in accordance with Specification MIL-I-16104.

3.5.4.25 Instrument transformers. - Instrument transformers where used shall conform to Specification MIL-I-1361.

3.5.5 Wiring and connections-

3.5.5.1 Harnessing. - All wiring shall be neatly formed into groups which shall be laced with noninflammable (glass) cord or nylon ties and supported or clamped in a manner which will prevent chafing of the insulation due to vibration and shock. The clamps shall be made of noninflammable material. Where metal is used, it shall be covered with a noninflammable material to prevent chafing of the wire insulation. There shall be no splices in wire and all connections shall be made at the terminal studs of the devices or at terminal blocks. The wire groups running from hinged panels shall be formed and clamped so that sharp bends do not occur with the panel in either the open or closed position.

3.5.5.2 Slack. - Sufficient slack shall be provided so as not to impair movement or put undue stresses on the wire or components in those places where movement of components may be expected. Slack shall also be provided to prevent undue stresses on terminal connections due to shock or vibration. Where wire terminals are used, sufficient slack shall be provided for at least two replacements of the part in the event that the wires are damaged or have to be clipped at the terminals during disassembly.

3.5.5.3 Mechanical supports. - All electrical connections shall be designed and provided with supports so as to prevent breakage and minimize changes in performance due to vibration, inclination or shock.

3.5.5.4 Cable entrance provisions. - Removable cable entrance plates shall be provided both top and bottom of each bulkhead or deck mounted control cabinet and on each side of motor generator mounted control cabinets, so that the necessary cables can be brought into the enclosure. Removable cable entrance plates mounted on the top or side of the control cabinet shall be provided with gaskets.

3.5.5.5 External cable connections. - External cable connections to the equipments shall terminate at terminal boards accessibility located near the cable entrance. All terminal boards for external connections shall be accessible from the front of the enclosure. Circuitous routing or wiring shall be avoided.

3.5.5.6 Insulation protection. - Wherever wires are run through holes in metal partitions, the holes shall be equipped with grommets for mechanical protection of insulation. Care shall be exercised in the running of wires to insure that they are not carried over or bent around any sharp corner or edge.

3.5.5.7 Studs. - Studs of potted parts shall be so designed and fastened to the insulating strip or plate (or the enclosure itself if this conforms to insulation requirements) as not to cause any degradation in the moisture excluding property of the enclosure by the normal soldering and resoldering of external leads to the studs.

3.5.5.8 Wire end connectors. - The ends of each wire (except for pigtail leads or parts requiring solder connections) shall be connected to the part studs or to terminal boards by means of solderless connectors of a type approved by the bureau or agency concerned or by forming a wire around the part stud and retaining the wire loop in a cup or crimped washer. The strands of the wire loop shall be secured together by soldering. All nuts, bolts and screws used for electrical connections shall be secured by a locking device, however that a locking device need not be provided where solderless connectors for conductors below 4000 circular mils are used. Connections utilizing quick disconnect connectors (in accordance with Specification MIL-C-5015) in conjunction with a positive locking arrangement, to prevent circuit opening under shock or vibration, may be used as approved by the bureau or agency concerned.

3.5.5.9 Soldering connections. - Pigtails (wire leads which are permanently affixed to a part), studs (soldering post which are permanently fixed to a part), soldering lugs (projections which solder to ends of wire to facilitate attachments to solderless connections), and soldering terminals (items which are attached to the chassis, or insulating boards, strips, or posts to serve as solder junctions for two or more wires or leads) come under this heading. All soldering studs, lugs, and terminals shall be notched or otherwise provided with means for mechanically securing the wire or lead prior to the application of solder. All solder terminations shall be completely plated with tin or silver. All solder type studs and terminals shall be sturdily designed and shall be constructed of materials which will render them suitable for repeated soldering and unsoldering operations without breaking or loosening.

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All solder type studs shall be mounted in such manner as to preclude their loosening or rotation due to soldering operations or from strains due to attached wires or leads. No more than three wires, including wires from parts, shall terminate at any one terminal.

3.5.5.10 Wire identification.- All wires shall be marked at each end by stamping the wire number on the wire terminal or on an insulated sleeve which shall be slipped over the wire close to the terminal. The sleeving shall fit tightly over the wire or over the round portion of the terminal to prevent the sleeving from sliding on the wire and to insulate the terminal. Sleeving shall be type F, grade A, form U, class II, category 1, white, in accordance with Specification MIL-I-631. Markings shall be permanently stamped on the sleeving in such a manner as to remain legible after repeated handling and exposure to grease.

3.5.5.11 Identification and information plates-

3.5.5.11.1 Identification plates.- Identification plates shall be furnished on each piece of control equipment which is separately mounted. When the control equipment is mounted in a common control panel, identification plates shall be furnished on the outside of the complete unit as well as on major individual components within the enclosure such as voltage regulators, frequency regulators, motor controllers, and so forth. The minimum data to be marked on the plate shall include the name of the unit, manufacturer's name, manufacturer's serial numbers Federal stock number, drawing number, contract number, year of manufacture, rating data, a blank space for the Government inspector's official acceptance stamp and identification to associate the equipment with its parent equipment (such as, for use with 10 kw. 230 DC/120AC 400 cycle MG SNSN).

3.5.5.11.2 Information plates.- The function of each control and indicator on the face of the panel as well as the function of each internal adjustment, connector and test point, within the enclosure shall be indicated by means of information plates adjacent to the device.

3.6 Repair parts.- Onboard repair parts, based on the total number of identical parts or sets of parts furnished for each ship on a contract or order, shall be supplied as specified in Table IX. This table includes parts for the complete motor generator set, including auxiliary devices such as the voltage regulator, speed governor, motor starter, and so forth, and takes precedence over repair part requirements in individual referenced specifications.

3.6.1 Irreplaceable parts.- If any part listed in Table IX forms part of an assembly or sub-assembly and is not normally a replaceable part of the assembly, then the complete assembly shall be furnished in lieu therof as a repair part.

3.6.2 Matched parts.- Where matching of parts is required in the equipment, the matched set of parts shall be considered one part and furnished as such if required as a repair part.

3.7 Drawings:

3.7.1 General requirement.- Drawings shall be in accordance with Specification MIL-D-963 and as specified herein.

3.7.2 Kind of drawings.- Drawings shall be class A in accordance with Specification MIL-D-963 and the following types:

(a) Preliminary drawings
(b) Final drawings

3.7.2.1 Preliminary drawings.- These drawings which shall be submitted for approval shall be either blue prints, blue line prints or equal.

3.7.2.2 Final drawings.- Final drawings shall be either microfilms, tracings, or direct reading blockline prints in accordance with Specification MIL-D-963. Where microfilms are furnished, one full size non-reproducible copy of each drawing shall also be furnished.

3.7.3 Drawings to be furnished.- The following drawings are required to be furnished:

(a) Motor-generator
(b) Control panel 1/
(c) Motor starter
(d) Voltage regulator
1/ (e) Frequency regulator
(f) Schematic
(g) Interconnection
(h) Installation

1/ The voltage regulator and frequency regulator drawings may be combined on a single drawing or the information may be included on the control panel drawings. When separately mounted control equipment is furnished in lieu of a complete control panel, a "Motor-generator power supply system" drawing shall be furnished in lieu of the control panel drawing. This drawing shall consist of a schematic diagram for the complete system conforming to 3.7.3.2 (f), an interconnection diagram for all components of the system conforming to 3.7.3.2 (d) and part component data as required by 3.7.3.2 (b) for all components for which separate master drawings are not furnished. Components which may be necessary to make a complete system but are not furnished by the contractor (such as circuit breakers, rheostats, and so forth) shall also be described and shown in the schematic and wiring diagrams and a note added to indicate that they were not furnished by the contractor.)

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Table IX - Quantities of onboard repair parts to be furnished per ship

Parts required	Identical parts, or sets[1/] of parts, installed per ship							
	1	2	3	4	5-8	9-20	21-50	50-100
Bearing, sets	1	1	1	1	2	3	4	5
Brush holders, complete with springs	1	1	1	1	2	3	4	5
Brush holder springs	1	2	3	3	4	6	7	8
Brush rigging insulation, sets	1	1	1	1	2	3	4	5
Brushes, sets	2	3	4	4	5	6	8	12
Capacitors ea. size & type	1	1	1	1	2	3	4	5
Circuit breaker trip element if AQB type breaker is furnished, or complete circuit breaker if ALB type is furnished	1	1	1	1	2	3	4	5
Contacts for rheostats and motor Controller (except for overload devices), sets	1	1	1	1	2	3	4	5
Overload heaters for motor controller, sets	1	1	1	1	2	3	4	5
Rectifiers, ea. size & type	1	2	3	3	4	6	7	8
Resistors, ea. size & type	1	1	1	1	2	3	4	5
Rheostats, ea. size & type (if springs and contacts are not replaceable)	1	1	1	1	2	3	4	5
Saturable reactors only if the reactors contain magnet wire of less than 1000 circular mils cross-sectional area complete set (or four, whichever is less)	1	1	1	1	1	2	3	4
Shunt coils for motor controller	1	1	1	1	1	2	3	4
Springs for rheostats and motor controller (except for overload devices), sets	1	1	1	1	2	3	4	5
Switches, ea. size & type	1	1	1	1	2	3	4	5
Transistors, ea. size & type	1	2	3	3	4	6	7	8
Voltage regulator diodes	1	2	3	3	4	6	7	8
Voltage reference diodes	1	2	3	3	4	6	7	8

[1/] A set is all parts of the same size and type installed in one motor generator.

3.7.3.1 Motor-generator drawing. - Information to be shown on the motor-generator master drawing shall be as follows:

(a) Descriptive data.

(1) Identification

a. Manufacturer's type, class or model number

(2) Reference specification

a. A statement as follows: "The D.C. / A.C. motor-generators shown hereon conforms to Specification MIL-M-19097____ dated ____ with the following exceptions: (State name or list all exceptions as applicable)."

(3) Guaranteed performance

a. Overall efficiencies at 1/4, 1/2, 3/4 and full load.

(4) Motor generator classification

a. Enclosure
b. Speed
c. Ambient temperature
d. Mounting
e. Duty
f. Insulation class

(5) Motor classification

a. Horsepower, voltage and range, current and range, class of insulation, type of winding and field current, at no load and rated load
b. Armature core diameter, core length and nominal airgap.
c. Number of main poles and commutating poles.
d. Commutator diameter, length, wearing depth, under cut, insulation between bars, insulation to ground and creepage to ground.

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e. Brush information consisting of manufacturer's grade, military grade, thickness, width, length, top and bottom bevel, number of brush studs and number of brushes per stud.

(6) Generator classification
a. KW rating, voltage, phase, power factor, frequency current, excitation current and voltage at no load and rated full load and insulation class.
b. Generator constants
1. Short circuit ratio (minimum calculated values only).
2. Transient reactance (maximum calculated values only).
3. Subtransient reactance (minimum calculated values only).
c. Stator core diameter, core length, nominal airgap
d. Rotor core diameter, core length
e. Number of collector rings, ring diameter, ring material, wearing depth, insulation to ground, creepage to ground
f. Brush information consisting of manufacturer's grade, military grade, thickness, width, length, top and bottom bevel, number of brushes per stud.

(7) Shaft, bearings and lubrication
a. Shaft material
b. Type of bearings
c. Type of lubrication
d. Note identifying the type and class of the applicable standard bearing puller and adapter (in accordance with Specification GGG-P-781).

(8) Weights
a. Weight of complete motor generator
b. Weight of rotor
c. WK^2 of the rotor (type II motor generators only)
d. Calculated Moment of Inertia about each of the three principle axis through the center of gravity (Type II motor generator only).

(b) D. C. motor winding data

(1) Armature winding data to include type of winding, number slots, number of coils per slot, turns per coil, pitch of coil, size of conductor insulation (specification and type) feet of wire per coil, weight of copper per coil, resistance of wound armature in ohms at 25°C.
(2) Armature, winding diagram, facing connection end.
(3) Cross sectional view of armature slot showing coils and all insulation
(4) Cross sectional view of field poles showing coils and all insulations.
(5) Dimensional plan and end view of each type of armature and field coils. Dimensions before and after forming shall be shown.
(6) Motor field winding data to include size and shape of conductors, conductor insulation (specification and type), turns per coil, feet of wire per coil, number of conductors in parallel and resistance per coil in ohms at 25°C.
(7) Field coil connection diagram
(8) Table of all insulation material indicating type of material form, thickness applicable government specification and indicate where used. Materials listed in this table need not be repeated in the basic list of materials.
(9) The following additional insulation data shall either be shown on each motor-generator master drawing or may be shown on a separate insulation procedure master drawing showing the various processes, procedures and methods used by the manufacturer. If the latter is selected, the motor-generator master drawing shall reference the applicable portions of the insulation procedure master drawing.
a. Sequence of all winding and insulating operations and applicable specifications for all material used.
b. Tape application including type, number of layers, amount of overlap, and so forth.
c. Sketches as necessary to completely describe all operations
d. Ground insulation, slot cell insulation, phase insulation, coil spacers and separators, end turn insulation, insulation under banding wire, support pads, distance that slot cell insulation extends beyond laminations, alternate materials slot wedges, method of bringing out and insulating leads from each coil, method of securing windings, and so forth.
e. Varnish treatment schedules including type of varnish, impregaration methods, time cycles, and baking temperatures for wound coils prior to installation as well as treatments for completed subassemblies.

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(c) A. C. generator winding data
(1) Same type of information as required for dc motor winding data (see 3.7.3.1(b)).

(d) Test reports
(1) The periodic test report number(s) and date used to obtain design approval shall be referenced on the drawing.

(e) Construction details

(1) An assembly drawing showing a side and two end views with all parts of the motor generator identified with piece numbers in the list of materials. The side view shall show a longitudinal section of the motor-generator above the centerline of the shaft. The end views shall show quarter sections (or more if necessary) of the motor and generator. These assembly views shall be drawn so that they will be legible when the drawing is reduced to technical manual size. The assembly views shall show all information as indicated herein. Separate detail views of subassemblies shall be provided where the sectional views do not clearly show the information required.
(2) Method of securing all rotating elements to the shaft. This includes bearing inner races, fans, commutators, collector rings, armatures, rotating field assemblies, spiders, spacers, sleeves, and so forth. The drawing shall clearly indicate the method used to prevent both axial and rotational movement of all items secured to the shaft. Where interference fits are used to prevent axial movement, the inner and outer diameters and tolerance, shall be indicated.
(3) Method of securing all stationary elements to the frame. Details furnished to be the same as indicated for 3.7.3.1(e) 2 above.
(4) Bearing housing construction, fits and tolerances. Bearing fits and tolerances may be indicated in tabular form.
(5) Baffles and screen guards and method used to attach them to the end brackets or frames.
(6) Method of attaching terminal box to frame
(7) Method of bringing cables out of frame into terminal box and method of protecting these cables against chafing at sharp metallic corners.
(8) Method of securing the motor and generator leads to the terminal boards in the terminal box.
(9) Overall dimensions of terminal box, and the number and size of cover bolts.
(10) A detailed working drawing of the shaft including, dimensions, finishes and tolerances.
(11) Sectional view of collector ring assembly showing direction of rotation, design of brush holders and brush holder rigging, including studs, and method of supporting the brush holder rigging, including studs, against excessive deflection due to high impact shock. The drawing of the brush holder shall show the basic design, method of adjusting spring pressure and method attaching brush shunt. The materials of the brush holder body, brush holder spring, and brush holder rigging shall be indicated in the List of Material.
(12) Direction of air flow through the frame
(13) Lifting means provided
(14) All overall dimensions including mounting and maintenance clearances.

3.7.3.2 Control equipment drawing. - The control panel, voltage regulator, and frequency regulator master drawings shall include the following information as applicable. The motor controller and field rheostat drawings shall include information as required by specifications MIL-C-2174 and MIL-R-15109 respectively.

a. Descriptive data for the equipment

(1) Type of equipment
(a) Control panel voltage regulator, frequency regulator
(2) Manufacturer's identification
(a) Type, class, model number
(3) Reference specification
(a) A statement that the equipment conforms to specification MIL-M-19097_____dated _____with the following exceptions: (State none or list exceptions)
(4) Equipment classification (as applicable)
(a) Type of equipment
(b) Enclosure
(c) Shock classification
(d) Ambient temperature
(e) Electrical ratings, voltage, current phase frequency, voltage, and so forth, as applicable.

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(f) Mounting
(g) Ventilation
(h) Weight of complete assembly or part where furnished separately
(i) Special features
(j) Insulation class

(5) Test reports
The periodic test report number or numbers and date used to obtain design approval shall be referenced on the drawing.

b. Parts design data

(1) Transformers, reactors and magnetic amplifiers-
(a) Core material, core or lamination form and dimensions (include small dimensioned sketch if necessary for clarity).
(b) Winding data including the number of turns, tape, wire size and type, terminal identification, d.c. resistance between terminals at 25°C, nominal voltage between terminals, volt - amp ratings for transformers, and current ratios for current transformers.
(c) Part designation as assigned in the elementary schematic diagram and part number used in the list of material.
(d) Complete material details including applicable specifications for materials used. Details of core boxes, layer and coil insulation, ground insulation, method of impregnation and treatment, potting methods and materials, terminal boards, grommets, leads, shall be included.

(2) Resistors.
Type, resistance, voltage rating, standard, Military type number, type of mounting type of terminals, type of adjusting band, and any other special data which may be necessary for ordering exact duplicates from the resistor manufacturer.

(3) Rheostasts.
Nominal diameter number of plates connection, voltage rating, current rating (both ends, number of steps and part designation.)

(4) Capacitors.
Type, capacitance, voltage rating, standard Military type number and part designation. Also indicate the maximum ac (peak and rms) and dc (average) voltages which will be applied to the capacitor in the application and the condition at which these maximum voltage occur (transient load conditions, 50 percent rated current etc.) If dry electrolytic capacitors are permitted (see 3.5.4.13) add the following drawing note:
"It is to be noted that these electrolytic capacitors deform and may become instable when stored or left idle for extended period of time."

(5) Current switches
a. Type, number of position, position marking, voltage rating, current rating, contact arrangement, number of poles, position tabulation and part designation.
b. Chart showing contact position for each switch

(6) Semi-conductor devices.
Each device shall be identified in the list of material by manufacturer's part or drawing number to permit direct ordering (JEDEC number is adequate). Also provide mounting details and application (not device capability) data as follows:
a. Rectifiers using silicone diodes.
1. Circuit symbols and piece number
2. Maximum reverse voltage peak and rms (complete rectifier circuit)
3. Maximum load current dc amperes (complete circuit)
4. Maximum diode case temperature at design ambient-use for diodes rated against core temperature.
5. Maximum ambient temperature directly adjacent to diodes-use for diodes rated against ambient temperature - no cooling plates.
6. Cooling plate size, thickness and material where so mounted.
7. Air velocity if forced cooled (linear feet per minute.)
8. Circuit (for example, single phase bridge, and so forth).
9. Total diodes in circuit
10. Diodes in series per arm
11. Diodes in parallel per arm
12. Description of insulation hardware if insulated from a cooling plate.
b. Rectifiers using selenium cells (in stacks or cartridges).
1. Circuit symbol (s) and piece number.
2. Maximum a.c. voltage applied to the rectifier circuit.
3. Maximum d.c. load current through rectifier circuit
4. Maximum cell temperature (shall not exceed 75 C.)

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5. Cell dimensions (3x3 inches, 5x6 inches, and so forth).
6. Air velocity if forced cooled (linear feet per minute)
7. Rectifier circuit (single phase bridge, and so forth).
8. Total stacks or cartidges used in circuit
9. Stack or cartridge circuit (if different from rectifier circuit
10. Cells in series per arm of stack or cartridge
11. Cells in parallel
12. Special requirements specified in the purchase order but not covered in specification MIL-R-18281 but considered necessary for this application (such as a tighter reverse current limit for magnetic amplifier or control application)

c. Voltage regulator diodes.
1. Circuit symbol (s) and piece number
2. Range of actual voltage levels which circuit will accept (with or without circuit adjustment from regulator diodes and still give required equipment performance _____volts to_____ volts
3. Regulation (including temperature error) permitted in an installed regulator diode under all conditions of operation (assume equipment is initially adjusted for optimum performance) ± _____mv.
4. Maximum level of steady-state average reverse current through regulator diode (steady-state shall be considered any time in excess of 30 seconds): _____ ma.
5. Maximum transient reverse current under any operating condition ____ ma for_____ sec duration and _____ frequency of occurrence
6. Maximum ambient temperature at diode (no cooling plate)
7. Maximum case temperature of diode (complete for diodes rated against case temperature)
8. Cooling plate size, thickness and material where used
9. Air velocity if forced cooled (linear feet per minute)
10. Total diodes in circuit
11. Diodes in series per circuit element (complete where more than one circuit element is used in the voltage regulator such as in a bridge type regulator circuit.)
12. Description of insulation hardware if insulated from a cooling plate

d. Voltage reference diodes (See regulator diodes also)
1. Circuit symbol (s) and piece number
2. Range of reference voltage levels which circuit will accept (with or without circuit adjustment) from the voltage reference diode(s): _____volts to _____ volts
3. Error permitted in an installed voltage reference diode under all conditions of operation: ± _____millivolts
4. Circuit design current through voltage reference
(a) Nominal or average value: _____ma
(b) Regulation about nominal value: ± _____ma
5. Maximum ambient temperature (include internal heat rise above outside design ambient): _____a
6. Number of voltage reference elements in series: _____(operation in parallel is not permitted)

e. Transistors
1. Circuit symbol(s) and piece number
2. Max. collector to emitter voltage _____ volts
3. Max. collector to base voltage _____ volts
4. Max. collector current _____ma.
5. Max. base current _____ma.
6. Maximum collector power dissipation _____mw.
7. Maximum ambient temperature (include internal heat rise at transistor at 50°C design ambient _____°C
8. Maximum core temperature of transistor at 50°C equipment design ambient and any condition of required equipment operation _____ C. (Where rating is based on case temp.)
9. Air velocity of forced cooled
10. Cooling plate size, thickness and material where so mounted
11. Description of insulating hardware if insulated from a cooling plate

(7) Meters.
Scale, range and red line rated marking

(8) Circuit breakers.
Type, frame size, thermal element rating, instantaneous trip calibration, and any special features which may have been provided.

(9) Motor controllers.
Type, size, protection features,

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overload heater rating, and any other special features which may have been provided or minor deviations from the master drawing for the motor starter.

f. Schematic diagram.
(1) A schematic diagram of the complete system to clearly represent the operation of the equipment. The following features shall be incorporated.
(a) Each part, such as resistors, capacitors, and so forth shall be identified by symbol (R-resistor, T-transformer, L-reactor, and so forth).
(b) In addition to the part designation, the numerical value of all resistances, capacitances and inductances shall be noted adjacent to the part.
(c) Identification of all terminals shall be shown.
(d) Simplicity in reading and understanding the operation of the system shall be stressed throughout.

g. Wiring diagram.
(1) A wiring diagram for each of the major subassemblies and interconnection of the subassemblies. The following features shall be incorporated in the drawing:
(a) All parts shall be shown in correct relation to physical location within a particular enclosure.
(b) Terminals of all assemblies and subassemblies shall be clearly shown and marked.
(c) Designation of parts shall conform to those assigned in the schematic diagram. Also, the piece numbers corresponding to those shown in the list of material shall be indicated.
(d) Wire size, type, applicable specifications, and type of terminals shall be shown in tabular form.
(e) Lead identification.
(f) Continuous lines to represent leads between components need not be drawn if the lead identification used clearly indicates the termination point of each lead.

h. Description of theory of operation.
(1) A brief description of theory of operation of the system to enable the approval authority to understand the circuitry.

i. Construction details
(1) Two representative views of each enclosure showing the following:
(a) Overall dimensions of the enclosure
(b) Drilling and mounting dimensions
(c) Outline of all parts, subassemblies and assemblies contained in each enclosure identified with item or piece number corresponding to item or piece number contained in the List of Material
(d) The parts need not be drawn in detail nor to scale but should represent general configuration of the part and show its general location in the enclosure
(e) Details of construction showing cable entrance provisions; door stop and positioning device; guard rails; captive bolts; gaskets; and other details as applicable.
(f) Outline drawing of parts or components furnished separately (not mounted in an enclosure) to show critical dimensions including mounting dimensions.
(g) Location of terminal busses for connecting armature cables (two views).

3.7.3.3 Installation drawings. - Data to be shown on these drawing shall be as follows:

(a) Dimensional outline of all major assemblies (motor generator, control panels, and so forth, overall and principle dimensions in sufficient detail to establish the limits of space in all directions required for installation, operation and servicing, exclusive of space required for personnel. Include the amount of clearance required to permit opening doors and hinged panels; clearances required for withdrawal of parts and sub-assemblies such as end brackets, rotating assemblies, and so forth, for the motor generator and chassis for the control equipment panels; clearances for cable access, and clearances for ventilation. Also indicate the location of meters and operating controls on control equipment panels so that the panels may be located at a convenient operating height.
(b) Weight of each enclosure and separately furnished assembly
(c) Center of gravity of all major assemblies such as motor generator, control panel, motor starter, and so forth.
(d) A block diagram of all major assemblies showing interconnection between these assemblies. Terminal identification, as well as maximum current rating of each cable should be shown.
(e) Moment of inertia for Type II motor generators.

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3.7.4 Technical manuals. - Two copies of technical manuals shall be packaged with each motor generator. Technical manuals shall conform to type II of Specification MIL-M-15071 and as follows:

(a) Front Material
- (1) Chapter 1 General Information
  - a. Section 1 General data
    Component List
    Characteristic of each component (motor generator, voltage regulator, frequency regulator, motor controller, and so forth).
  - b. Section 2 Introduction
    Brief explanation of what it is, where it is used, what it will do. Block diagram showing major components of the system and brief explanation of the function of each component in the system.
- (2) Chapter 2 Detailed description
  - a. Section 1 Motor generator
    Detailed description of major parts or assembly with photograph.
  - b. Section 2 Voltage regulator
    Block diagram of assemblies (voltage reference, bias, mag amp), and explanation of each, with photograph (door closed)
  - c. Section 3 Speed governor
    Block diagram of assemblies (speed reference bias mag amp and so forth), and explanation of each, with photograph (door closed)
- (3) Chapter 3 Principle of operation
  - a. Section 1 Voltage regulator
    Schematic diagram of various assemblies of the voltage regulation with explanation of function of each part.
  - b. Section 2 Frequency regulator
    Schematic diagram of each assembly of the speed governor with explanation of function of each part.
- (4) Chapter 4 Operating instruction
  - a. Section 1 Safety precautions
  - b. Section 2 Start up
  - c. Section 3 Manual operation
  - d. Section 4 Automatic operation
  - e. Section 5 Parallel operation
  - f. Section 6 Shut down
- (5) Chapter 5 Installation
  - a. Section 1 Installation instruction
  - b. Section 2 Check out before start up
- (6) Chapter 6 Maintenance
  - a. Section 1 Preventive maintenance
    Routine inspection and care of equipment
  - b. Section 2 Corrective maintenance
    Open loop test
    Trouble shooting charts
    Tables showing points of measurements of voltage, current, resistance, and so forth
    Adjustments after repair
- (7) Chapter 7 Parts identification
  - a. Section 1 Introduction
  - b. Section 2 Parts tabulation. Photograph of controls, door open with components indicated by arrows and symbols.
- (8) Chapter 8 Test Data
  - a. Section 1 Heat runs, voltage and speed regulation parallel operation, harmonic content, no load saturation curve transient response, resistance at 25°C insulation resistance cold, short circuit current.

(b) Drawings

(c) Appendix
- (1) Motor controller insert
- (2) Circuit breaker insert
- (3) Rheostat insert

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 In-process inspection. - The supplier shall perform in-process inspection of processes, materials, parts, subassemblies, assemblies and finished machines to demonstrate conformance with applicable drawings and specifications. The supplier's in-process inspection shall include the applicable items of inspection contained in Appendix I

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herein, selected in accordance with the design, manufacturing methods and sequence of operations. In-process inspection procedures, measuring gauges, instruments and written records shall be acceptable to the inspector. The supplier shall have a quality control system in accordance with the requirements of Specification MIL-Q-9858.

4.2.1 Performance tests.- Performance tests specified herein shall be conducted by the supplier using facilities, methods and instrumentation acceptable to the inspector.

4.2.1.1 Test reporting.- Test results shall be recorded by the supplier on test forms prepared by the supplier and approved by the inspector. Provision shall be made for recording of all required data. Each test report form shall be identified by the number of this specification and by the particular test requirement paragraph number and its title. When oscillograms are required, each trace shall be identified as to the circuit measurement involved and its calibration shall be indicated to permit ready use without reference to associated documents. Noise test shall be reported in accordance with Specification MIL-E-22843 and as follows:

(a) Type I motor generators - short report
(b) Type II motor generators - Periodic test, detailed report
Routine test, short report

4.2.2 Verification inspection.- The inspector may perform complete or partial inspection on an appropriate sampling basis to verify that the requirements of this specification have been met and to substantiate that the manufacturer's inspection records indicate the true quality of the motor generators and associated control equipment.

4.3 Sampling

4.3.1 Sampling procedures for routine tests.- All motor generators on the contract or order shall be subjected to the routine tests specified in Table X. The tests shown therein shall be performed on completely assembled units.

4.3.2 Sampling procedure for periodic tests.- The first motor generator power supply of each design and size offered for delivery shall be subjected to the periodic tests specified in table X. Thereafter one periodic test on identical equipment will be required every 12 months. A periodic test shall be required after any change in design which affects the performance characteristics. If routine test data reveals variations beyond a normal manufacturing tolerance, the inspector may require that any or all of the periodic tests be made on a particular motor generator to demonstrate that it conforms to this specification. If motor generators and controls do not meet performance requirements under periodic tests, that motor generator and those of the same design and lot shall be rejected. After the design has been modified and the revisions incorporated into the sample, the sample may be resubmitted for tests.

4.4 Test reports.- All test data shall be approved by the inspector prior to the shipment of the equipment. After approval, one file copy of all periodic test data shall be forwarded to the bureau or agency concerned. This data shall be bound into a 9 by 11-1/2 inch binder. The cover and title page shall give sufficient information to identify the equipment. A table of contents shall be included and list each test required by table X. No special test format will be required. However, each test shall be prefaced by the name of the test and applicable test paragraph as indicated in table X. This data submitted shall be copies of the actual data taken on the test floor and not retyped data. The forms used shall allow sufficient columns to make instrument corrections and necessary calculations. The inspector shall sign the cover sheet of the test report and note thereon the number of test sheets verified.

4.5 Special provisions-

4.5.1 Periodic tests.- The shock test, weight measurement and externally excited vibration tests need only be conducted on the first unit of a given design.

4.5.2 Routine tests.- Routine tests which require that the motor generator and controls be operated together may be conducted as follows:

(a) Use one motor generator to test all control units.
(b) Use one set of controls to test all motor generators.
(c) Test any motor generator and any set of controls together.

The motor generator and control which is used for the complete periodic test shall not be used in the routine testing of any of the other units.

4.6 Methods of tests.- Test procedures shall be as specified herein.

4.6.1 Creepage and clearance.- Creepage and clearance distances between current carrying parts in the motor generator and control equipment shall be measured to determine compliance with Specification MIL-E-917.

4.6.2 Enclosure.- Acceptability of the enclosure shall be determined by visual examination or measurements to determine compliance with 3.3.5 and 6.1.

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Table X - Routine and periodic tests

Description of test	Applicable test paragraph		Test to be conducted while motor-generators and control equipment are operating together as a complete system	Test which may be conducted on motor-generator alone	Test which may be conducted on control equipment alone
	Routine tests	Periodic tests			
Creepage and clearance	4.6.1	4.6.1		X	X
Enclosure	4.6.2	4.6.2		X	X
Weight		4.6.3		X	X
Air gap	4.6.4	4.6.4		X	
Resistance at 25°C.	4.6.5	4.6.5		X	
Insulation resistance (cold)	4.6.6	4.6.6		X	X
Dielectric	4.6.7	4.6.7		X	X
Airborne noise:					
Type I motor generator	4.6.8.1.2	4.6.8.1.1		X	
Type II motor generator	4.6.8.2.2	4.6.8.2.2			
Commutation	4.6.9	4.6.9	X		
End play	4.6.10	4.6.10		X	
Inclined operation		4.6.11		X	
Overspeed	4.6.12	4.6.12		X	
Externally excited Structureborne noise:					
Type I motor generator	4.6.13.1.1	4.6.13.1.1		X	
Type II motor generator	4.6.13.2.2	4.6.13.2.1			
Externally excited vibration		4.6.14			
Inherent speed (frequency) regulation		4.6.15	X		
No load saturation curve for generator		4.6.16		X	
Synchronous impedance Curve		4.6.17		X	
Shock		4.6.18	X		
Regulated frequency adjustment range and steps		4.6.19	X		
Regulated voltage adjustment range and steps		4.6.20	X		
Manual frequency adjustment range and steps		4.6.21	X		
Manual voltage adjustment range and steps		4.6.22	X		
Automatic frequency and voltage regulation	4.6.23.2	4.6.23.1	X		
Transient response of frequency regulator (deviation and recovery time)	4.6.24	4.6.24	X		
Transient response of voltage regulator (deviation and recovery time)	4.6.25	4.6.25	X		
Parallel operation (kw and kvar sharing)		4.6.26	X		

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Table X - Routine and periodic tests (Cont'd.)

Description of test	Applicable test paragraph		Test to be conducted while motor-generators and control equipment are operating together as a complete system	Test which may be conducted on motor-generator alone	Test which may be conducted on control equipment alone
	Routine tests	Periodic tests			
Short circuit characteristics		4.6.27	X		
Overspeed protection					
Type II motor generator	4.6.28	4.6.28	X		
Rated load heating		4.6.29.2	X		
Special heating tests					
Type II motor generator (Only)		4.6.29.3	X		
Resistance (hot)		4.6.5			
Insulation resistance (hot)		4.6.30	X		
Efficiency at 4/4, 3/4, 2/4 and 1/4 load		4.6.31	X		
Harmonic analysis		4.6.32	X		
Amplitude modulation		4.6.33	X		
Line unbalance (3 phase generator only)		4.6.34	X		
Voltage balance (3 phase generator only)		4.6.35	X		
Component ratings	4.6.36			X	X
Repair part		4.6.37			

4.6.3 Weight. - The weight of the complete motor generator, complete rotating element of the motor generator, complete control panel (or each individual control equipment enclosure and switchboard mounted components in the case of separately mounted control equipment) shall be measured and recorded (see 3.4.3).

4.6.4 Air gap measurement. - The motor generator stationary air-gaps shall be measured and recorded on adjustable poles only to determine compliance with 3.4.16. All measurements shall be taken from iron to iron and as close as possible to the center of the pole face.

4.6.5 Resistance (cold and hot). - The resistance of all motor and generator windings shall be measured when the equipment is at room temperature. The resistance and temperature of the windings shall be measured after the rated load heat run (4.6.29) and the calculated temperature for this resistance shall be recorded adjacent to the cold resistances on the test report.

4.6.6 Insulation resistance (cold). - This test shall be made before the dielectric tests. Prior to the application of the test voltage, all circuits shall be thoroughly discharged by shorting them to the frame and the motor generator may be disconnected from the control equipment. The test voltage shall be applied between all electrically isolated circuits (motor generator as well as control equipment) and between each circuit and ground. Circuit diagrams should be carefully studied prior to conducting this test to ascertain that circuits which may be isolated by transformer are not inadvertently neglected. When testing between the circuits and ground, all circuits of the control equipment and the motor generator may be tied together so that only one test voltage need be applied to each, the motor generator and the control equipment providing that the insulation resistance when tested in this manner meets the minimum value specified in 3.2.29. Insulation resistance shall be measured with an insulation resistance indicating instrument. The time of test voltage application shall be not less than 60 seconds. The temperature of the circuits at the time of the test

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shall be measured and recorded. Insulation resistance measurements shall be corrected to 25°C. Unless otherwise specified in the contract or order, corrections shall be made on the basis of insulation resistance doubling for each 15°C. decrease in temperature. The relative humidity at the time of the test shall be measured and recorded.

4.6.7 Dielectric tests. -

4.6.7.1 General. - The dielectric test shall be made after all other tests have been completed. The dielectric test shall be made on the completely assembled motor generator and control equipment and not upon individual parts within the motor generator or control equipment. An exception is made in the case of repair parts which require dielectric tests.

4.6.7.2 Points of application. - The test voltages of a magnitude specified in 3.2.31 shall be successively applied between each electric circuit and all other electric circuits and metal parts. The test voltage shall be applied in such a manner as to preclude the possibility of pitting the bearings in case of insulation failure. Voltage need not be applied between stationary and rotating windings of the motor generator.

4.6.8 Airborne noise

4.6.8.1 Type I motor generators (rotating equipment only). -

4.6.8.1.1 By instrument (for periodic test). - The airborne noise measurements shall be made and recorded in accordance with Specification MIL-E-22843 with the motor generator operating at rated load and nominal input voltage to determine compliance with 3.3.12.

4.6.8.1.2 By observation (for routine tests). - Observation shall be made of motor generator noise during the progress of testing to determine that the motor generator is free from any degree of noise comparably greater than that inherent in the given type and size of motor generator. If any objectionable noise is detected the inspector may insist on a test by instrument as specified in 4.6.8.1.1 to ascertain that the unit meets the requirements of 3.3.12.

4.6.8.2 Type II motor generators and associated controls. -

4.6.8.2.1 By instrument (for periodic test). - The airborne noise measurements shall be made and recorded for motor generators and associated control equipment in accordance with Specification MIL-E-22843 for the following operating conditions:

Input voltage	Load
210 volts d.c.	100 percent
210 volts d.c.	50 percent
355 volts d.c.	100 percent
355 volts d.c.	50 percent

4.6.8.2.2 By instrument (routine test). - Airborne noise measurements shall be made and recorded for motor generators and associated control equipment in accordance with Specification MIL-E-22843 for the one condition which produced the highest reading during periodic test in 4.6.8.2.1.

4.6.9 Commutation. - Inspection shall be made to determine that commutation is satisfactory. The use of shunts across the commutating pole field terminals is not acceptable. After the brush position has been set, it shall not be changed during the progress of the remaining tests. The brush position shall be set prior to conducting any of the performance tests which may be affected by the brush positioning. Commutation shall be checked at no load, 25 percent rated load, 50 percent rated load, 75 percent rated load, and rated load for maximum and minimum motor input voltages specified in 3.2.2. Commutation observed at all load points up to and including rated load shall be completely black (no sparks showing). Any sparking at steady loads (not during transient periods) up to rated load shall be cause for rejection of the equipment. After completion of all tests, the commutator and brushes shall be inspected for signs of excessive wear, pitting or other injuries and such defects shall be cause for rejection of the equipment.

4.6.10 End play. - The end play of all motor generator rotating elements shall be measured to determine compliance with 3.4.7.4.

4.6.11 Inclined operation. - The motor generator and control equipment shall be tested for inclined operation. The angle of inclination shall be as follows:

(a) Type II motor generators and control equipment: Motor generator shaft inclined 45 degrees, generator end low. Motor generator shaft inclined 45 degrees, motor end low. Control equipment inclined 45 degrees in any direction.

(b) All other motor generators and control equipment: Motor generator shaft inclined 15 degrees, generator end low. Motor generator shaft inclined 15 degrees, motor end low. Control equipment inclined 15 degrees in any direction.

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The motor generator and control equipment shall be run at no load for a period of not less than 1 hour for submarine equipment and 30 minutes for all other equipment at each of the positions of inclination as specified herein. The control equipment may control the motor generator during the test or be tested separately under simulated no load loading. During the progress of these tests it shall be ascertained that there is no apparent overheating of any component, dropping out of circuit breakers and the mechanical balance is as good as it was in the normal horizontal position, that there is no pounding or grinding at the bearings, and that the lubrication is satisfactory.

4.6.12 Overspeed test. - Each motor generator shall be operated at 125 percent rated speed for a period of not less than 5 minutes. A check shall be made for noise, balance, and smoothness of running during the test, for loose solder, and for evidence of distortion, injury or noticeable change in the condition of the commutator bars, balance weights, fans, windings, or any other parts after shutdown.

4.6.13 Structureborne noise. -

4.6.13.1 Type I motor generators. -

4.6.13.1.1 Internally excited vibration test. - Each motor generator shall be tested to determine compliance with 3.4.4.2. This test shall be conducted after the motor generator has been subjected to the 125 percent, 5 minute overspeed test specified in 4.6.12. The unit shall be mounted elastically at a natural frequency less than one-quarter of the rotational frequency of unit. To accomplish this, the minimum static deflection of the mounting should be determined as shown on figure 1. In no case shall the deflection exceed one-half the original height of the elastic element. Measurements shall be made on the bearing housing and in the direction giving the maximum vibration when the motor generator is operating at rated speed and at rated load. Test data submitted for each machine shall include a description of the test set up including compression of elastic mount, instruments used, location of readings taken, actual instrument readings, and any corrections or calculations made. This test shall also be run as part of the post shock tests of the unit which was HI shock tested in order to ascertain that the HI shock test has not affected the unit in such a manner that the equipment does not meet the post shock internally excited vibration limits of 4.6.16.b(6). Failure to meet the internally excited vibration limit after shock shall be considered as cause for rejection of the equipment.

4.6.13.2 Type II motor generators and control equipment.

4.6.13.2.1 By instrument (for periodic test). - The structureborne noise measurements shall be made and recorded for motor generators and associated control equipment in accordance with Specification MIL-E-22843 for the following operating conditions:

Input voltage	Load
210 volts d.c.	100 percent
210 volts d.c.	50 percent
355 volts d.c.	100 percent
355 volts d.c.	50 percent

4.6.13.2.2 By instrument (routine test). - Structureborne noise measurements shall be made and recorded for motor generators and associated control equipment in accordance with Specification MIL-E-22843 for the one condition which produced the highest reading during periodic test in 4.6.13.2.1.

4.6.14 Externally excited vibration test. - All motor generator controls shall be tested in accordance with MIL-STD-167 to demonstrate that the equipment meets the requirement of 3.2.10.

4.6.15 Inherent speed regulation. - With the controls under manual operation the following data shall be recorded for each condition to demonstrate that the equipment meets the requirements of 3.4.6, motor voltage, generator KW load, generator voltage and generator frequency. With nominal voltage applied to the motor (see 3.2.2) and no load rated voltage on the generator, adjust the speed or frequency to rated value. The frequency and voltage adjustments shall not be changed for the remainder of the test. Readings shall be taken for each of the following conditions, in any order.

Motor voltage (see 3.2.2)	Generator load
Nominal	0
Maximum	0
Maximum	50 percent Unity power factor (pf)
Maximum	100 percent Unity pf
Minimum	50 percent Unity pf
Minimum	100 percent Unity pf
Minimum	0

4.6.16 Generator no load saturation curve. - Sufficient data to plot a no load saturation curve for the generator shall be obtained. A copy of this curve shall be included in the test records. Points for this curve shall be taken as follows: Four below 60 percent of rated voltage; one between 60 and 90

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percent; one at rated voltage; and three between rated voltage and full field excitation. Readings shall always be taken with increasing values of field current and when it is necessary to reduce the field, it must be reduced to zero and then increased to the desired value. Data recorded for each point shall consist of field voltages and currents, speed generator voltage (each of the 3 phase voltages and average of the 3 phase voltages in the case of 3 phase generators), motor input voltage and motor input current. The motor generator shall be operating at rated speed and with rated voltage applied to the motor during this test.

4.6.17 Synchronous impedance curve. - Sufficient data to plot a synchronous impedance curve for the generator shall be obtained. A copy of this curve, which may be plotted on the same sheets as the saturation curve, shall also be included in the test records. Six readings from 150 percent rated current shall be recorded.

4.6.18 High impact shock test. - The test shall be as specified for type A of Specification MIL-S-901 and as specified herein:

(a) General. - The motor generator and control equipment shall be energized during the test and may be tested simultaneously or separately. The frequency regulator and voltage regulators and static exciter may be tested when operating with their intended motor generator or under simulated loading. All equipment or parts shall be mounted on the shock machine in a manner simulating the actual installation on board ship. The weight designation of the shock test shall be as required by the combined weight of the equipment being tested. The output of the equipment shall be recorded before and after each blow to determine compliance with 4.6.18(b). If simulated loading is used the values recorded shall be within the limits required to meet the regulation requirements when operating as a complete power supply.

(b) Failure to perform principal function. - Any of the following failures shall be considered as cause for rejection of the equipment:

(1) Breakage of any parts.
(2) Appreciable distortion or dislocation of any parts such as mounting feet, poles, coils, brushes, brush holders, brush holder studs and bearings.
(3) A value of insulation resistance of the motor generator or any of the control equipment (corrected to 25°C) less than that permitted by 3.2.29.
(4) Objectionable noise. Any noticable increase or decrease in noise of the motor generator when operating at rated speed should be investigated and its cause determined. However, if disassembly is required for this check, it should be made during the test specified in 4.6.8.
(5) Low dielectric strength. The motor generator and control equipment shall be operated to obtain approximately rated temperature rise of iron and windings, then disconnected from its load and its source of power and dielectric strength test made to check the condition of the insulation. This dielectric strength test shall be made in accordance with 4.6.7.2 except that it shall be made with an applied voltage equal to 65 percent of that specified.
(6) Failure to meet twice the internally excited vibration limit specified in 3.4.4.2 after completion of the shock test.
(7) A shift in regulated voltage or frequency of more than plus or minus 0.25 percent. The shock test on the voltage regulator and speed governor shall be conducted while they are controlling the motor generator at any convenient load. The speed of the motor and the generator voltage, or average of the 3 phase voltages in the case of 3 phase generators shall be recorded before and after each blow. No adjustments or change in input or output shall be made between blows.
(8) Failure to meet the frequency regulation and voltage regulation test specified in 4.6.23.2 after completion of the shock test.
(9) The equipment shall be operated at full load for 2 hours and its overall performance checked. Any noticeable increase in heating, or other abnormal operation noted and investigated to determine its cause.
(10) Failure to pass inspection. After all the checks and tests 4.6.18(b)(1) through (9) have been conducted, the motor generator and control equipment shall be disassembled and thoroughly inspected for damage. The extent of disassembly need be only to the point where the condition of the various components can be easily observed.

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(c) Disposal of shock-tested equipment.-

(1) Equipment which has been subjected to the high impact shock test and has failed to perform any of the principle functions specified herein will not be acceptable, either in whole or any of the parts, until it has been modified and successfully passed the HI shock test.

(2) Equipment which has been subjected to the high impact shock test and has successfully performed all the principle functions specified herein will be acceptable under a contract or order providing that ball bearings are replaced in the motor generator.

(d) The place at which the stock test shall be conducted.- Shock tests may be performed at the contractor's plant, commercial laboratory, or Government laboratory which is suitably equipped to perform these tests. In either case, the tests are to be conducted at the expense of the contractor. Where stock tests are conducted at a Government laboratory, copies of all master drawings shall accompany the equipment.

(e) The number of generators to be tested.- Unless otherwise specified in the contract or order, one motor generator of the longest core length in each diameter and of similar construction shall be shock tested. If the manufacturer desires, he may submit a motor generator frame of less than the longest core length for the shock test. However, satisfactory compliance with the shock requirements will only determine conformance of motor generators having a core length equal to or less than the core length of the motor generator tested.

(f) Shock extension.-

(1) Motor generator shock test approval may be extended from the longest core length (same outside diameter and similar construction) to a motor generator of a shorter core length. The manufacturer shall submit a letter of request together with the drawings of each unit. Upon receipt of the bureau or agency approval, the manufacturer shall note on the master plan the test number and date of original shock test, the inspector's approval letter, serial number and date, and a statement of shock extension including bureau or agency approval letter.

(2) The extension of shock test from one control panel and voltage regulator, or both, to another shall be based on comparison of size, weight, and component distribution. The manufacturer shall submit a letter of request together with the drawings of each unit. Drawing notation shall be as specified in 4.6.18(f)(1).

4.6.19 Regulated frequency adjustment range and steps.- The tests for compliance with 3.2.20 shall be conducted as follows:

(a) With rated frequency applied to the motor and the motor generator operating at no load under automatic operation.

(1) Record the maximum and minimum frequency adjustment.

(2) Record the frequency for three consecutive steps for at least five points over the range specified.

(b) Repeat (1) and (2) above for rated load at rated power factor.

(c) With the maximum and minimum input voltages specified in 3.2.2 applied to the equipment, the maximum and minimum limits of adjustability shall be recorded.

4.6.20 Regulated voltage adjustment range and steps.- The tests for compliance with 3.2.18 shall be conducted as follows:

(a) With rated voltage applied to the motor and the motor generator operating at no load under automatic operation.

(1) Record the maximum and minimum voltage adjustment.

(2) Record the voltage for three consecutive steps for at least five points over the range specified.

(b) Repeat (1) and (2) above for rated load at rated power factor.

(c) With the maximum and minimum input voltages specified in 3.2.2 applied to the equipment. The maximum and minimum limits of adjustability shall be recorded.

4.6.21 Manual frequency adjustment range and steps.- The test for compliance with the manual frequency adjustment limits specified in 3.2.21 shall be conducted as specified in 4.6.19 except that the controls shall be under manual operation.

4.6.22 Manual voltage adjustment range and steps.- The test for compliance with the manual voltage adjustment limits specified in 3.2.19

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shall be conducted as specified in 4.6.20 except that the controls shall be under manual operation.

4.6.23 Automatic frequency and voltage regulation.-

4.6.23.1 Periodic test.-

(a) The test shall be started with the rotating and control equipment at room temperature.

(b) Start the equipment with nominal voltage applied to the motor and with no load on the generator. Adjust the average voltage to rated and the frequency to rated; no further adjustment shall be made to the automatic voltage and frequency adjusting control or any other part of the regulators during the remainder of this test. The first set of readings shall be taken within 2 minutes after the equipment is energized. The generator voltage and current (or each of generator phase voltages, average of the phase voltages, and each of the three output line currents in the case of three phase generators), KW output, speed, frequency (may be computed from speed if suitable meter is not available), generator field voltage and current, motor input voltage and current motor main and auxiliary field voltage and current shall be the minimum data recorded for each set of readings taken during the tests.

(c) Apply full load to the generator and permit the equipment to reach its full load temperature rise before proceeding with the remaining parts of this test. A set of readings shall be taken every 30 minutes during warmup.

(d) A set of readings shall be recorded for each of the following tests specified in table XI:

Table XI - Automatic voltage and frequency regulator test conditions.

Percent of rated kw load	Power factor of load (lagging)	Motor input voltage (see 3.2.2)	Test no.
0	—	Min. (minimum)	1
		Nom.(nominal)	2
		Max.(maximum)	3
25	0.8	Min.	4
		Nom.	5
		Max.	6
	1.0	Min.	7
		Nom.	8
		Max.	9

Table XI - Automatic voltage and frequency regulator test conditions. (Cont'd)

Percent of rated kw load	Power factor of load (lagging)	Motor input voltage (see 3.2.2)	Test no.
50	0.8	Min.	10
		Nom.	11
		Max.	12
	1.0	Min.	13
		Nom.	14
		Max.	15
75	0.8	Min.	16
		Nom.	17
		Max.	18
	1.0	Min.	19
		Nom.	20
		Max.	21
100	0.8	Min.	22
		Nom.	23
		Max.	24
	1.0	Min.	25
		Nom.	26
		Max.	27
75	0.8	Min.	28
		Nom.	29
		Max.	30
	1.0	Min.	31
		Nom.	32
		Max.	33
50	0.8	Min.	34
		Nom.	35
		Max.	36
	1.0	Min.	37
		Nom.	38
		Max.	39
25	0.8	Min.	40
		Nom.	41
		Max.	42
	1.0	Min.	43
		Nom.	44
		Max.	45
0	—	Min.	46
		Nom.	47
		Max.	48

Load	Generator terminal connection	Power factor of load (lagging)	Motor voltage input (see 3.2.2)	Test no. 1/
Single Phase 15 Percent Rated Current	$T_1 - T_2$	0.8	Nominal	49
"	$T_1 - T_2$	1.0	Nominal	50

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Table XI - Automatic voltage and frequency regulator test conditions. (Cont'd.)

Load	Generator terminal connection	Power factor of load (lagging)	Motor voltage input (see 3.2.2)	Test no. 1/
Rated Current (Cont'd)	$T_2 - T_3$	0.8	Nominal	51
"	$T_2 - T_3$	1.0	Nominal	52
"	$T_3 - T_1$	0.8	Nominal	53
"	$T_3 - T_1$	1.0	Nominal	54

1/ Tests 49 through 54 are applicable to three phase generators only. The generator shall be loaded with a 3 phase, 75 percent rated, 0.8 power factor load before the single phase 20 percent rated current load is applied during tests 49 through 54.

The manufacturer may conduct any number of tests, in any sequence for a KW load. The tests must be conducted in order of ascending and descending load as shown in table XI.

Tests 49 through 54 may be taken before or after tests 1 through 48 or interspersed with tests 16 through 21 and tests 28 through 33.

(e) All voltages or the average phase voltages in the case of three phase generators recorded during the above tests shall be within the limits specified in 3.2.14.

(f) All frequencies recorded during the above tests shall be within the limits specified in 3.2.16.

4.6.23.2 Routine test.- This test shall be conducted in accordance with 4.6.23.1, except that only the following tests will be conducted in the order listed

Test No.
3
22
47
17
23
41
51
15

4.6.24 Frequency deviation and recovery time.- Frequency deviation and recovery time shall be as follows:

(a) With the generator operating at no load and with rated voltage applied to the motor, the load shall be increased in one step to 50 percent load at 0.8 power factor.

(b) With the generator operating at 100 percent load at 0.8 factor and with rated voltage applied to the motor, the load shall be reduced in one step to 50 percent load at 0.8 power factor. Oscillograms shall be taken to determine compliance with 3.2.17. This test may be conducted concurrent with the automatic voltage and speed regulation test 4.6.23.

4.6.25 Voltage deviation and recovery time.- With the generator operating at no load and with rated voltage applied to the motor 50 percent load at 0.8 power factor (single phase for single phase generators and balanced 3 phase for 3 phase generators) shall be suddenly applied in one step. This test shall be repeated removing this same load from a generator which was initially carrying rated load at rated power factor. Oscillograms shall be taken to determine compliance with 3.2.15. The voltage trace shall be adjusted so as to have no less than 3 inches peak to peak displacement for rated voltage. The film speed shall be such as to give approximately 8 cycles per inch, and sufficient film used to record continuous operation from at least 0.5 second before the instant of load application or removal to at least 4 seconds of operation after the instant of application or removal of load. This test may be conducted concurrent with the automatic frequency and voltage regulation test 4.6.23.

4.6.26 Parallel operation.- The parallel operation test shall be conducted with each motor generator connected to the same or different power supplies, and the speed and voltage regulators in automatic operation, adjust the control equipment so that rated voltage and frequency is obtained and the KW and KVA loads of each genrator are equal for a combined load of 25 percent. No further adjustment of the control equipment (either motor or generator controls) shall be made for the remainder of the test. The total load shall then be varied from 25 percent to 100 percent of combined KW and KVA rating of the parallel generators at approximately rated power factor in three approximately equal steps and back to 25 percent load in the same manner. For submarine motor genrator power supplies, the test shall be repeated at the maximum and minimum input voltages specified (see 3.2.2). The actual difference in KW and KVAR expressed in percentage of the KVA rating of one power supply shall be computed and recorded.

4.6.27 Short circuit characteristics.- The sustained short circuit current shall be measured by means of an ammeter to determine compliance with 3.2.27. The contractor shall also demostrate that the equipment can withstand the sustained short circuit current for a period of two minutes. When circuit breakers are furnished by the contractor, this test shall be conducted with the circuit breaker locked

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in (or by passed if it is of a type which cannot be locked in) so as to prevent tripping before the 2 minutes. There shall be no evidence of smoking, burning, or burning odor as a result of this 2 minute test. When circuit breakers are furnished by the contractor, this test shall be repeated (after the equipment has cooled down) to demonstrate that the circuit breaker will trip in less then 2 minutes. The actual time shall be recorded. The voltage shall automatically return to within the regulation band limits when the short circuit is removed. Oscillograms of the fault current and voltage shall be taken.

4.6.28 Overspeed protection (type II motor generator only). - The performance of the overspeed protection shall be demonstrated and shall meet the requirements of 3.5.4.9. The actual speed at which the motor generator shuts down shall be recorded. The periodic test report shall include a brief statement describing the manner in which the test was performed.

4.6.29 Heating tests-

4.6.29.1 General.- All heating tests on the motor generator and control equipment shall be conducted at the same time. Simulated heating tests will not be permitted on the control equipment or motor generator. All heating tests shall be made on a production model. In those cases where the loading of control circuit components may be greater at some low or intermediate load point than it is at rated load or overload, additional tests will be required to demonstrate that these components are not overheating at these loads.

4.6.29.2 Rated load heating test.- The equipment shall be operated at rated load at rated power factor and with nominal voltage applied to the motor until the temperature has stabilized in all components. However, in no case shall the duration of the test be less than 4 hours. During the first 3 hours, temperature measurements shall be made and recorded at 30 minutes intervals. After that readings shall be made and recorded at 15 minute intervals. It shall be considered that stabilized temperatures have been reached, when at least four consecutive readings taken at 15 minutes intervals do not differ by more than 2°C. After shutdown, temperature rises of the rotating and stationary elements shall be determined in accordance with 4.6.29.5.3 and 4.6.29.5.4. Maximum temperature rises recorded shall be within the limits specified in table III.

4.6.29.3 Special heating tests for type II motor generators and control equipment.- All type II motor generators and associated control equipment shall receive a rated load heating test for each input voltage specified in 3.2.2 in addition to the heating tests specified in 4.6.29.2. The procedures of the test shall be the same as required for the full load heating test 4.6.29.2. Temperature rises shall not exceed the rated load limits specified in Table III. If this test is conducted immediately following (15 minutes or less) the rated load heating test (4.6.29.2). The four hour minimum time limit does not apply.

4.6.29.4 Data to be recorded.- The following data shall be the minimum recorded for each reading taken during the heating tests:

(a) Motor input voltage and current.
(b) Output power (measured on load side of circuit breaker)
(c) Output phase voltages and line currents.
(d) Power factor.
(e) Room ambient.
(f) Air temperature at a minimum of four locations within the control equipment enclosures.
(g) Temperature of air entering and leaving all enclosures.
(h) Thermometer or thermocouple readings for all parts

The temperature rise of all parts shall be determined by one of the methods specified in table III; however, the temperature rise of all motor and generator windings which have a resistance of 1.0 ohm or more shall be determined by the rise in resistance method. Where shutdown temperature of rotating elements (for windings less than 1 ohm) are taken in accordance with 4.6.29.5.3., extrapolation curves need not be included in the test report.

4.6.29.5 Measurement methods.- Measurement methods used in the heating tests shall be as specified herein. Detailed procedures of method I (thermometer) and method II (resistance) shall be as specified in Specification MIL-E-917. In determining temperature rise, no correction shall be made for barometric pressure, humidity or for any differences in heat transfer characteristics between the test ambient temperature and the specified 50°C. ambient temperature.

4.6.29.5.1 Ambient temperature.- The ambient temperature shall be measured in accordance with Specification MIL-E-917.

4.6.29.5.2 Potted components.- The temperature rise of all encapsulated or potted components shall be determined by embedding thermocouples in the samples to be used for this test.

4.6.29.5.3 Rotating elements.- The temperature rise of commutators, collector rings, motor

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armature windings, and generator rotating field or armature windings shall be determined as follows: On shutdown, preheated thermometers (preheated to a temperature approximately equivalent to the temperature rise permitted for the applicable part plus the room ambient at the time) shall be placed on the respective parts in accordance with method I procedures of Specification MIL-E-917, where applicable. Every precaution shall be taken to reduce to a minimum the period of time elapsing between the stopping of the machine and the determination of the temperature. A curve should be plotted with temperature readings as ordinates and time as abscissa. The curve should be extrapolated back to the instant of shutdown. The temperature at the instant of shutdown shall be recorded and the temperature rise determined. This rise shall be in accordance with the limits specified in table III. In addition, the temperature rise of all rotating and station windings which have a resistance greater than 1.0 ohm shall be determined by method II of Specification MIL-E-917.

4.6.29.5.4 Stationary elements of motor generator.- The temperature rise of motor and generator stationary coils laminations or field poles and the outer races of both bearings shall be measured by method I of Specification MIL-E-917 where applicable, the temperature rise of all coils which have a resistance greater than 1.0 ohms shall be measured by method II of Specification MIL-E-917 after shutdown.

4.6.29.5.5 Qualified products items (QPL).- QPL items such as rheostats, resistors, circuit breakers, motor starters, capacitors, instruments, and instrument transformers shall have been previously tested and are satisfactory when used within their rating and provided there is no excessive rise in air temperature within the complete control equipment enclosure. When all of the control equipment is integrated within a single enclosure, the rise of all transformer coils, motor controller coils, resistors, rheostats, case temperatures of capacitors, and air temperature within 1/2 inch of all instruments shall also be measured by method I of Specification MIL-E-917 during the heat runs and be within the limits specified in table III.

4.6.29.5.6 Other components.- The temperature rise of components not covered by 4.6.29.5.2 through 4.6.29.5.5 shall be measured as follows:

(a) Transformer windings, magnetic amplifier windings, and so forth by method I of Specification MIL-E-917.

(b) Semiconductor devices - The ambient temperature of semiconductor devices shall be measured by method I of Specification MIL-E-917 within 0.5 inches of the device on the low temperature side (side from which air starts passage over device). The case or cell temperature required for selenium rectifiers and other semiconductor devices relying upon a cooling fin or other surface to dissipate heat from the case to the ambient shall be measured by placing a thermocouple in direct contact with the case (on center cell near through bolt or stud of a selenium rectifier stack). Position on the case shall be on the high temperature side of the case. In no instance shall the thermocouple be placed on the cooling fin or other heat dissipating surface. The selenium cell or plate is an exception since the rectifing junction is distributed across the cell.

4.6.30 Insulation resistance (hot).- Immediately following the heat run, measurements of insulation resistance shall be taken. (For details of test see 4.6.6).

4.6.31 Efficiency.- The efficiency of the motor generator power supply shall be measured in automatic operation for 1/4, 1/2, 3/4, and full load. Motor input voltage, motor armature current, total input current, calculated KW. input, field voltages and currents for all motor and generator fields, generator voltage (each phase and average for 3 phase generators), generator line currents measured at the gnerator terminals, output line current measured at the load side of the circuit breaker, load power factor KW output, generator frequency, and the calculated overall efficiency shall be the minimum data recorded for each load point. A detailed breakdown of losses will not be required. However, if an analysis is made for the contractor's own design files, it is requested that a copy of this breakdown of losses be included in the test report. The data taken during the automatic frequency and voltage regulation test specified in 4.6.23 may be used to compute the efficiencies.

4.6.32 Harmonic analysis.- A harmonic analysis shall be made with the equipment operating at no load, and full load unity power factor. All harmonics up to the 40th harmonic shall be recorded. Harmonics of less than 0.1 percent may be disregarded in the determination of the total operation when the harmonic analysis is made.

4.6.33 Amplitude modulation.- Amplitude modulation of the generator output voltage shall be measured to determine compliance with 3.2.23. This test shall be conducted with the controls in automatic with the generator voltage and frequency at rated value. The percentage of amplitude modulation shall be recorded for no load and full load unity power factor.

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4.6.34 Line unbalance (3 phase generators only).- The data recorded during test 49 through 54 of 4.6.23 shall be used to determine compliance with the line unbalance requirements specified in 3.2.24. The maximum difference in phase voltages for each of the load points shall be computed and recorded. The voltage values used shall be measured at the output terminals of the power supply (load side of generator circuit breaker) and not at the generator terminals.

4.6.35 Voltage balance (3 phase generators only).- The data recorded during tests 1 through 48 of of the automatic frequency and voltage regulation test of 4.6.23 shall be used to determine compliance with the voltage balance requirements specified in 3.2.25. The maximum difference in line voltages for each of the load points shall be computed and recorded. The voltage values used shall be measured at the output terminals of the power supply (load side of the generator circuit breaker) and not the generator terminals.

4.6.36 Component ratings.- The inspector shall ascertain that the rating of all components such as resistors, rectifiers, capacitors, transformers, instruments, rheostats, and so forth, is as shown on the approved master drawing.

4.6.37 Repair parts.- Repair parts shall be subjected to the following tests as applicable:

(a) Insulation resistance.
(b) Dielectric.
(c) Creepage and clearance.
(d) Materials.
(e) Reistance at 25°C.

5. PREPARATION FOR DELIVERY

5.1 General.- The basic equipment and repair parts shall be prepared for delivery for the following shipments as specified (see 6.1). Type II motor generators shall be prepared for shipment in accordance with Specification MIL-E-22845.

(a) Domestic shipment and early equipment installation and for storage of onboard repair parts.
(b) Domestic shipment and storage or overseas shipment requirements.

5.2 Domestic shipment and early equipment installation and for storage of onboard repair parts.

5.2.1 Basic equipment.-

5.2.1.1 Preservation and packing.- Preservation and packing shall protect the equipment against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.2.1.2 Packing.- Packing shall be acceptable to the common carrier and afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.2.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2.2 Onboard repair parts.- Repair parts shall be preserved, packaged and marked in accordance with domestic shipment and storage requirements of Specification MIL-E-17555. No reusable boxes are required.

5.3 Domestic shipment and storage on overseas shipment requirements.-

5.3.1 Basic equipment repair parts.-

5.3.1.1 Preservation and packaging.- The basic equipment and repair parts shall be preserved and packaged Level A in accordance with Specification MIL-E-17555.

5.3.1.2 Packing.- The basic equipment and repair parts shall be packed by Level A or B in accordance with Specification MIL-E-17555. Reusable boxes are not required for packing of repair parts.

5.4 Technical publications.- Technical manuals shall be prepared for delivery in accordance with Specification MIL-E-17555.

5.5 Drawings.- Drawings shall be packed in accordance with Specification MIL-D-963.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of motor generators to be furnished (see 1.2).
(c) Rating (see 3.2.1).
(d) Input voltage type I only (see 3.2.2).
(e) Output voltage (see 3.2.3).
(f) Phase (see 3.2.4).
(g) Output frequency (see 3.2.5).
(h) Enclosures (see 3.3.5).
(1) Type of cooler if totally enclosed water cooled enclosure is specified (see 3.3.5.3).

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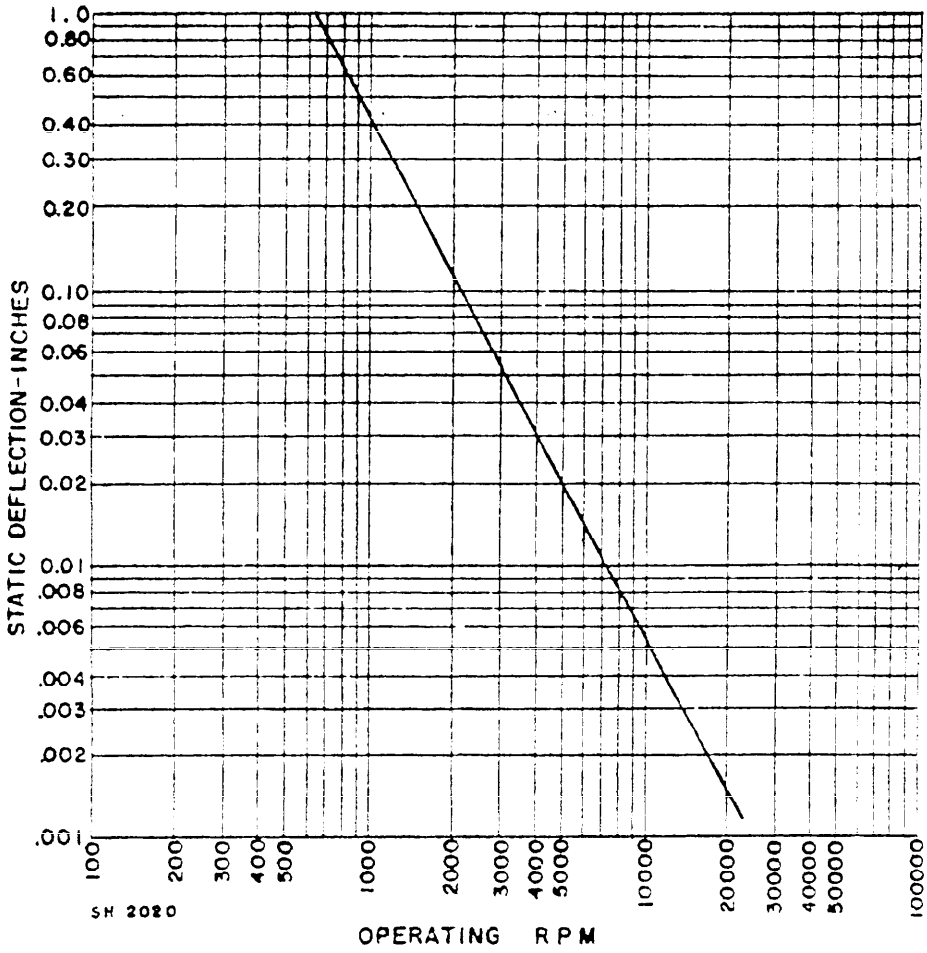
(i) Airborne and structureborne noise grade - Type II motor generator (see 3.3.13)
(j) Dimension and weight of Type II motor generators (see 3.4.3).
(k) Control arrangement (see 3.5.2).
(l) Dimensions and weight of control equipment (see 3.5.3.1).
(m) Mounting of control equipment (see 3.5.3.2).
(n) Preservation packaging, packing or marking requirements other than those required by paragraph 5.1 (see 5.2).

Notice - When government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise an in any manner licensing the holder or any person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Static Deflection-Inches



Operating RPM

Figure 1 - Static deflection of elastic mounting for internally excited vibration test.

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APPENDIX I

10. INSPECTION CHECK LIST FOR IN-PROCESS INSPECTION

10.1 This appendix lists those items which should be included in the in-process inspection of motor generators power supplies including control equipment furnished in accordance with this specification. This list is not an all inclusive list and additions and deletions as applicable can be made by the manufacturer subject to the approval of the inspector.

20. APPLICABLE DOCUMENTS

20.1 The following document of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-917 - Electric Power Equipment, Basic Requirements For (Naval Shipboard Use).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 To facilitate the in-process inspection procedure the manufacturer shall prepare a manufacturing flow chart showing the normal flow of material, parts, sub-assemblies, and assemblies through the manufacturing process. The manufacturer shall indicate on the flow chart the points at which the manufacturer conducts in-process inspection and these points shall be designated as inspection stations.

30.1.1 For each inspection station the manufacturer shall prepare an inspection check sheet listing the items inspected at that station. Where appropriate, a standard sketch shall be included in the check sheet showing dimensions to be checked and the allowable tolerances, finish requirements, etc. Space shall be left on the sketch or on the check sheet to record the results of dimensional, visual or other inspection.

A column shall be included on the check sheet for the initials or stamp of the manufacturer's representative conducting the inspection. A space shall be left on the inspection check sheet for the inspector's initials or stamp.

The inspection check sheet shall be suitably protected against damage or staining and will be physically attached to the item being inspected. When the component is assembled, the inspection check sheet for the component shall be attached to the assembly or otherwise made available

The words "approved drawing" as used herein mean a required drawing, such as Type I drawing, which is approved by the cognizant approval activity, or a manufacturer's shop drawing if needed to depict details for in-process inspection not shown on a required drawing. Manufacturer's shop drawings which agree with approved required drawings may also be used for in-process inspection. Inspection requirements are not to be waived or reduced by selection of drawings.

30.2 General inspection items. - General inspection item shall be as follows:

(a) Material is as specified on approved drawing. Material was ordered in accordance with applicable material or component specification and was inspected in accordance with the requirements of the material or component specification.

(b) All welding is done by qualified welders and is in accordance with the approved drawing. There is no evidence of non-fusion, weld cracks, under-size welds, incomplete welds, heavy porosity, weld splatter or slag.

(c) All soldered connections are solidly bonded; there is no cold soldering, no rosin joints, no corrosive flux, no fractured joints, or excess solder. Satisfactory connections were made prior to soldering. Bolted connections include approved locking devices and are secured against vibration. Solderless connectors are properly crimped and the connector and crimping tool are proper size.

(d) Finished castings are as shown on approved drawing and are clean and free of molding sand, cracks, blow holes, splits and deformations. Sufficient material is allowed for machining. Casting defects have not been covered by unauthorized repairs.

(e) Machining is as shown on approved drawing. The surfaces, including mating surfaces, as applicable, are smooth, square and are free of burrs, sharp edges, tool marks, chatter marks, and scratches and damage due to handling. Surface finish is as shown on approved drawing, and there are not tool marks except those normally associated with the indicated surface finish. There are no flaws exposed as the result of maching.

(f) No parts are government surplus or have been previously used or reclaimed.

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(g) All items including hardware (nuts, bolts, lockwashers, and so forth), are made of corrosion resistant material or are given a corrosion resistant treatment as shown on the approved drawing.
(h) The Government inspector may require a coil or winding to be cut apart to see the extent of varnish treatment and filling if there is any question as to the effectiveness of varnish treatment used.
(i) All bolts, nuts, set screws and other fasteners are secured in a manner which will preclude loosening in service. All locking devices are as shown on the approved drawing.
(k) Insulation creepage and clearance distances are in accordance with approved drawing. Creepage distances are not achieved by means of cemented or butt joints.

30.3 Motor generators

30.3.1 Items to be inspected prior to assembly. - Items to be inspected prior to assembly shall be as follows:

(a) Shafts.

(1) Items (a), (b), (e) and (f) of 30.2 apply.
(2) All dimensions, tolerances, keyways, fillets, shoulders, and surface finish are as shown on the approved drawing.
(3) Eccentricity tolerances including out of round for the bearing seats, bearing shoulder, collector ring seats, and the seats for the rotating armatures or rotors are as shown on the approved drawing.
(4) After welding on appurtenances, shaft has been stress relieved as shown on the approved drawing.
(5) The shaft was not built up in any way to correct errors or reclaim material (unless specific case basis approval of the bureau or agency concerned is obtained in advance).
(6) All shoulders are square as shown on the approved drawing.

(b) Collector rings.

(1) Items (a), (b), (d), (e), (f), and (j) of 30.2 apply.
(2) All dimensions and dimensional tolerances of rings, sleeve and sleeve insulation are as shown on the approved drawing.
(3) Manufacturing and assembly processes used are exactly as specified on the approved drawings. Specific temperatures and curing cycles are as shown on the approved drawing.
(4) Brush contact surfaces are smooth, free of sharp edges, burrs, porosity, deep tool marks and chatter marks. Slip ring grooves have no burrs.
(5) Terminal studs are fixed to the rings in a positive manner as shown on the approved drawing.
(6) Terminal stud insulation is as shown on the approved drawing.

(c) Commutators.

(1) Items (a), (c), (e), (f) and (j) of 30.2 above apply.
(2) The fit of components is in accordance with the approved drawing.
(3) Commutators have received a curing treatment in accordance with the approved drawing.
(4) Riser slots have no thin walls and are as shown on the approved drawing.

(d) Fans.

(1) Items (a), (b), (d), (e), (f), (g), (h), (i), and (j) of 30.2 apply.
(2) Fans, except those covered by 30.3.1(d)(5) below are balanced.
(3) All dimensions and the number of blades are as shown on the approved drawing.
(4) The fan surfaces which move the air are free from any irregularities, surplus weld material or any other projections which may be a source of airborne noise.
(5) Fans which break down into several parts are indexed in a manner such that they can be assembled in one way only.

(e) End brackets and bearing housings, end caps and frames.

(1) Items (a), (b), (e) and (f) of 30.2 apply.
(2) Sufficient metal is available for drilling and tapping.
(3) All drilling, tapping and bolt centers are as shown on the approved drawing. Holes are clean, free of chips and are drilled straight. There are no burred threads. Holes are spot faced and are located so that edge distance is adequate in accordance with the approved drawing.

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(4) All rust and grease is removed before painting. End brackets and frames fabricated and stored for periods prior to assembly were cleaned, and given a coat of primer or rust preventive before storing.
(5) The end brackets, except mating surfaces, such as bearing housings and rabbets, are cleaned, primed and painted on the inside surface prior to assembly. The inside of the frame is primed and painted if it did not receive at least two dips and bakes during the varnish treatment of the stationary electrical components.
(6) Mating surfaces of brackets and frame are concentric as shown on the approved drawing.
(7) All dimensions, dimensional tolerances and concentricities are as shown on the approved drawing.
(8) Bearing housings have sufficient metal to permit redrilling and bushing of the housing.

30.3.3 Items to be installed during assembly. -

(a) General.

(1) Items specified in 30.2 apply.
(2) All parts are interchangeable and no selective matching of parts is necessary.
(3) Assembly and disassembly of all equipment can be done without the use of special tools except as otherwise specified in the contract. Special tools are defined as those tools not listed in the Navy Stock List of General Stores. (Copies of this stock list may be consulted in the office of the Government inspector.)

(b) Items to be checked during winding and assembly of rotating elements.

(1) Keys as shown on the approved drawing are used to prevent rotational movement of all rotating parts, (such as collector rings, commutators, spiders, rotors or armatures).
(2) When rotating parts are pressed on the shaft, the pressure required to press on these parts is as shown on the approved drawing.
(3) When rotating parts are shrink fitted on the shaft, the interference shrink fits of the parts are as shown on the approved drawing.
(4) Axial movement of all parts is prevented by the methods shown on the approved drawing.
(5) Fans are secured to the shaft as shown on the approved drawing.
(6) Coil connections for rotating armature, AC generators and all equalizer connections are insulated adequately as shown on the drawing.
(7) Bearings are of the size and type as shown on the approved drawing.
(8) The inner races of ball and roller bearings are secured to the shaft by means of shaft shoulders, lock washers and nuts or by the opposed shoulder method as shown on the approved drawing.
(9) Lamination size and stacking are as shown on the approved drawing.
(10) Slot or ground insulation are as shown on the approved drawing.
(11) Slot or ground insulation extends beyond stacking as shown on the approved drawing.
(12) Wire size and type used is as shown on the approved drawing.
(13) Preformed coils formed in accordance with the dimensions shown on the approved drawing and given at least one varnish treatment with clear baking varnish before installation in the slots.
(14) Undue force is not required to put slot wedge in place and insulation is not damaged or pushed out of place as the wedge is inserted.
(15) Slot wedges are of material and size as shown on the approved drawing. Length of wedge exceeds length of slot as shown on approved drawing.
(16) Coils are not loose after the wedges are in place.
(17) The wedge size is proper for the size and shape of the slot and there is no possibility that the wedge will cock in the slot and slip out.
(18) Coil connections are adequately insulated as shown on the drawing.
(19) Coil support and phase insulation is as shown on the approved drawing. Phase insulation is inserted between the phase coils and is shaped to fit coil configuration.
(20) Coil extensions are insulated and secured as shown on the approved drawing.
(21) Insulation materials used are as shown on the approved drawing.
(22) Dimensions of wound coils are as shown on the approved drawing.
(23) Rotating field poles are secured as shown on the approved drawing.

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(24) If steel wedges are used to secure rotating field poles, the wedges are securely driven and positive assurance is provided that they will not work loose.

(25) Insulation is cleaned off and unless self fluxing wire is used coil wire ends are tinned before making soldered connections. Connections are properly soldered. There is no evidence of cold solder joints. Commutator riser slots are filled with solder and any excess removed.

(26) Lead wires are of the type shown on the approved drawing.

(27) Windings are mechanically secured as shown on the approved drawing.

(28) When preformed coils are used, the coils are given one varnish treatment prior to insertion in the slots or on the poles and the completed winding assembly, including winding and pole is given the varnish treatments with approved clear baking varnish. In the case where preformed coils are not used the complete assembly is given at least three varnish treatments.

(29) Type of varnish, treatment and baking time cycle, and baking temperatures are as shown on the approved drawing.

(30) The treated windings and coils are clean, smooth and glossy with good bonding and filling. Varnish seals are complete and show no signs of cracks or breaks. The completed windings have no air bubbles, air pockets, voids or dry spots on the surface and are not soft or sticky.

(31) There is no sign of excessive varnish buildup on one side of the winding assembly and lack of varnish buildup on other side. The thickness of the varnish on the winding assembly shall be uniform over the entire surface of the windings.

(32) Lead wires are insulated from ground and secured to prevent them from moving due to centrifugal force. Length and arrangement of wires shall permit ready repair; there is no aimless wiring resulting in "rats nests". Wiring agrees with the approved drawing.

(33) No glyptal or non-approved type of varnish or paint was put on any of the rotating elements.

(34) Coils or windings are not nicked or damaged during handling and processing.

(35) All completed rotating elements (including bearings), are dynamically balanced. The type of balance weights and method of securing are as shown on the approved drawing. They are secured in a manner such that they will not loosen in service. Balance weights are not attached in the airstream of the fan.

(36) Mica is under cut to specified depth. Copper segments are not damaged.

(c) Items to be checked during winding and assembly of stationary elements.

(1) Items 3.4(b), (10) thru (23) inclusive, (26) thru (32) inclusive, (34) and (35).

(2) Complete stator assembly or stator core is secured to frame or frame spider as shown on approved drawing.

(3) Force required to press complete stator assembly or stator core into the frame or frame spider is as shown on the approved drawing.

(4) Axial movement of the stator within the frame is precluded by means shown on the approved drawing.

(5) All lead wires are insulated from ground and secured within the frame with a suitable clamp or fastening device.

(6) All lead wires pass through the frame, enter the terminal box and are secured in such a manner to prevent chafing or abrasion as shown on the approved drawing.

(7) Terminal lugs of the type shown on the approved drawing are provided on the leads. Terminal lugs are properly crimped and there are no cut wire strands.

(8) Terminal boxes are of the type and size shown and are secured in the manner shown on the approved drawing.

(9) All leads are properly marked as shown on the approved drawing. It is possible to differentiate motor leads from generator leads.

(10) All nuts, bolts, screws and other hardware are tight and provided with locking devices as shown on the approved drawing.

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(11) Insulation material as shown on the approved drawing is placed between field coil and frame prior to securing the pole.
(12) Field pole coil is not unduly stressed while tightening pole securing bolts. Insulation is not damaged in the tightening process.
(13) Each field coil received at least three separate dips and bakes with an approved clear baking varnish either before or after assembly into the stator.
(14) All salient field coil connections were either soldered or bolted and a solid joint was made.
(15) All salient field coil connections are adequately insulated as shown on the approved drawing.
(16) Terminal board material is as shown on the approved drawing.
(17) Terminal studs are secured to the board in a manner to preclude turning when attaching external cables.

30.3.3 Items to be inspected during final assembly - Items to be inspected during final assembly shall be as follows:

(a) Items (a), (b), (c), (d), (e), (f), (g), (h), (i) and (j) of 30.2.
(b) End brackets properly match frames bearing housings and end caps. All holes align, there is no excess clearance, and no undue force is required to assemble parts.
(c) Coil ends do not protrude to a point where they may be damaged by assembly or removal of the end bracket and there is no possibility of interference with the rotating elements.
(d) Bearing outer races are secured in their housings as shown on approved drawing. The inside diameter of the retaining shoulders is within the dimensional tolerances shown on the approved drawing.
(e) There is no evidence of grease leakage past the close clearance non-rubbing seals into the motor generator or along the shaft.
(f) Spring washers, if used to provide preloading of the bearing, shall be selected and secured in the housing as shown on approved drawing.
(g) Seals of pre-lubricated bearings are not damaged and there is no evidence of grease leaking out of the bearing.
(h) Brush holder studs are secured as shown on the approved drawing.
(i) Brush rigging insulation is as shown on the approved drawing.
(j) Brush holders and springs are of the type and size shown on the drawing.
(k) Brush holders are secured as shown on the approved drawing.
(l) Brushes are of the manufacturer's grade designation shown on the approved drawing.
(m) Brushes have been properly seated on commutator or slip rings over a minimum of 90 percent of contact area of brush and excess carbon has been removed from the machine.
(n) Brush shunts are attached to the brush material by imbedding or riveting.
(o) Brush size is as shown on the approved drawing and fits properly in the holder without sanding or filing and brushes move freely in the brush holders in the direction parallel to the length of the brush.
(p) Brush tension is as shown on the approved drawing. Spring tension can be readily measured by spring scale.
(q) Brushes are properly aligned parallel to the shaft and do not extend beyond the edge of the slip rings or commutator.
(r) Brush lug terminals are properly secured by screws which are not used for securing the brush holders to the rigging.
(s) Brushes are properly spaced and staggered on the motor so that the commutator will wear evenly.
(t) Brush leads are furnished with lugs and are adequately secured.
(u) Air gaps have been adjusted to design value and are uniform within the limits shown on the approved drawing.
(v) Bearings have been lubricated with grease as specified on the approved drawing.
(w) Air baffles are securely attached and do not interfere with rotating elements.
(x) Lifting means is provided as shown on the approved drawing.
(y) Equipment enclosure is as shown on the approved drawing.
(z) Drain plugs are provided as shown on approved drawing.
(aa) Entire exterior, other than identification plates or shaft extensions is painted in accordance with Specification MIL-E-917.
(bb) Identification and instruction plates are as shown on the approved drawing; contain all required information and are legible.
(cc) Mounting feet are flat, square, and are as shown on approved drawing.
(dd) The overall dimensions, mounting dimensions, and location of terminal boxes are as shown on the approved drawing.
(ee) Friction type (rubbing) seals, are the type and are secured as shown on the approved drawing.
(ff) Where high tensile bolts or screws such as socket head types, are used, provision

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is made to prevent replacement with a lower tensile strength bolt or screw. There are no slotted head bolts or screws used to secure end brackets to the frame. Thread-cutting screws (self tapping) are not used to secure any part of the motor generator.

(gg) Spot facing is as specified in Specification MIL-E-917.

(hh) The motor generator, particularly the bearing housings, is clean and free of dirt, metal chips or other foreign materials.

(ii) The weight of the complete unit is as shown on the drawing.

(jj) Stationary lead wires do not contact rotating parts.

30.4 Control equipment

30.4.1 Items to be inspected on major parts.

(1) The following items were ordered in accordance with the applicable specification as modified by the contract requirements; are of the size, type and rating shown on the drawing; and are QPL items.

(a) Resistors and rheostats.
(b) Instruments, instrument transformers, shunts and other associated accessories.
(c) Motor starters and associated push-buttons.
(d) Circuit breakers.
(e) Capacitors.

(2) The following items are not necessarily QPL items but are of the size, type and rating shown on the drawing.

(a) Switches.
(b) Rectifiers.
(c) Molded terminal boards.
(d) All laminated materials such as glass mica glass (GMG), glass silicon glass (GSG); used for terminal boards; sub-panels; and so forth.

(3) All materials used in the construction of reactors, magnetic amplifiers, and transformers other than instrument transformers are as shown on the approved drawing and comply with the material requirements as shown on the approved drawing.

(4) All windings of reactors, magnetic amplifiers, and transformers other than instrument transformers were given a minimum of three varnish treatments or one vacuum impregnation and bake followed by one dip and bake with an approved clear baking varnish in accordance with the procedures shown on the approved drawing.

(5) No black varnish, glyptol, paint, or other treatment was given to the insulation of reactors, magnetic amplifiers, and transformers other than instrument transformers after the last dip and bake with clear baking varnish.

(6) The resistance of all windings of all reactors, magnetic amplifiers, and transformers other than instrument transformers is as shown on the approved drawing. Ratio of turns is as specified on approved drawing.

(7) The equipment enclosures are as shown on the approved drawing.

30.4.2 Items to be inspected on completed unit.

(1) All components are mounted as shown on the approved drawing.

(2) All parts which require servicing, repair, replacement, or periodic adjustment during the life of the equipment are readily accessible.

(3) All components are marked with identifying symbols and these symbols agree with those used on the approved drawing.

(4) All terminals of all components and terminal boards are marked with identifying letters and numbers, or both, and this identification agrees with those used on the approved drawing.

(5) Wire used for interconnecting components is of the type and size shown on the approved drawing.

(6) All wiring is neatly formed into groups and laced with noninflammable glass cord and supported or clamped in a manner which will prevent chafing of the insulation due to vibration and shock.

(7) Cable clamps used are of non-inflammable material.

(8) There are no splices in any of the wires.

(9) Wire groups from hinged doors and panels are formed and clamped in a manner that sharp bends do not occur with the panel or door in either the open or closed position.

(10) Sufficient slack in wiring is allowed so that the weight of the harness is not supported by the terminal connections and so that at least two replacements can be made if lugs are clipped off at the wire end.

(11) Cable entrance provisions are as shown on the approved drawing.

(12) Sufficient space is provided to bring external cables through these cable entrances, support them within the enclosure, and make connections at the appropriate terminals.

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(13) Wherever wires run through holes in metal partitions or chassis, grommets are provided for mechanical protection.
(14) Wires are not bent around sharp corners which may injure the insulation.
(15) All wires are connected by either bolted or soldered connections.
(16) All bolted connections are provided with locking devices.
(17) The soldered connections do not rely on the solder for mechanical strength. The wire was mechanically secured before soldering and the solder was used only to get a good electrical path.
(18) No more than 3 wires are soldered to any one terminal.
(19) Both ends of all wires are marked with designations as shown on the approved drawing.
(20) Hinged doors and panels do not bind when opening.
(21) A door stop and positioning device is furnished and functions satisfactorily.
(22) All identification and instruction plates are furnished as shown on the approved drawing.
(23) All information on identification and instruction plates is legible.
(24) A copy of the schematic and wiring diagram for the complete system as well as a copy of the operating instructions has been laminated in plastic and is stowed in a convenient location within the control panel.
(25) The location of the plastic laminated drawings and instructions is such that it does not interfere with convection air flow which may increase heating of components. In those cases where the location is questionable, heat runs of the control panels are run with the diagrams in place.
(26) The overall and mounting dimensions of the equipment are as shown on the approved drawing.
(27) The overall weight of the equipment is as shown on the approved drawing.
(28) The exterior and interior of the enclosure has been painted as shown on the approved drawing.
(29) All controls operate smoothly throughout their range.

APPENDIX II

10. SILICON RECTIFIER DIODES, SILICON VOLTAGE REFERENCE ELEMENTS, VOLTAGE REGULATOR DEVICES AND TRANSISTORS.

10.1 This appendix lists a selection of silicon rectifier diodes, silicon voltage reference elements, regulator devices and transistors. This list is not an all inclusive list and additions and delections as applicable can be made by the manufacturer subject to the approval of the inspector.

20. APPLICABLE DOCUMENTS

20.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY
MIL-E-1 - Electron Tubes and Crystal Rectifiers.
MIL-S-19500 - Semiconductor Devices, General specification for.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. SILICON RECTIFIER DIODES.

30.1 Silicon rectifier diodes shall be selected in accordance with table I.

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Table I - Silicon rectifier diodes

JEDEC (EIA) No.	Maximum ratings up to temperature specified				Required derating at higher temp. MA/°C	Maximum oper. temp.		Detailed specification
	V_R 1/	I_F	T_A	T_C		T_A	T_C	
1N483B	70	125	75	-	1.0	150	-	MIL-S-19500/118
1N485B	180	125	75	-	1.0	150	-	MIL-S-19500/118
1N486B	225	125	75	-	1.0	150	-	MIL-S-19500/118
1N645	225	250	75	-	2.5	150	-	MIL-E-1/1143
1N646	300	250	75	-	2.5	150	-	MIL-E-1/1143
1N647	400	250	75	-	2.5	150	-	MIL-E-1/1143
1N648	500	250	75	-	2.5	150	-	MIL-E-1/1143
1N649	600	250	75	-	2.5	150	-	MIL-E-1/1143
1N538	200	500	85	-	4.0	150	-	MIL-E-1/1084
1N540	400	500	85	-	4.0	150	-	MIL-E-1/1085
1N547	600	500	85	-	4.0	150	-	MIL-E-1/1083
1N253	95	1.0A	-	135	73	-	150	MIL-E-1/1024
1N254	190	400	-	135	29	-	150	MIL-E-1/989
1N255	380	400	-	135	29	-	150	MIL-E-1/990
1N256	570	200	-	135	15	-	150	MIL-E-1/991
1N1124A	200	2.0A	-	100	20	-	150	MIL-S-19500/104
1N1126A	400	2.0A	-	100	20	-	150	MIL-S-19500/104
1N1128A	600	2.0A	-	100	20	-	150	MIL-S-19500/104
1N1614	200	4.5A	-	135	75	-	175	MIL-E-1/1240
1N1615	400	4.5A	-	135	75	-	175	MIL-E-1/1241
1N1616	600	4.5A	-	135	75	-	175	MIL-E-1/1242
1N1199	50	12A	-	150	240	-	175	MIL-E-1/1108
1N1200	100	12A	-	150	240	-	175	MIL-E-1/1108
1N1201	150	12A	-	150	240	-	175	MIL-E-1/1108
1N1202	200	12A	-	150	240	-	175	MIL-E-1/1108
1N1203	300	12A	-	150	240	-	175	MIL-E-1/1108
1N1204	400	12A	-	150	240	-	175	MIL-E-1/1108
1N1205	500	12A	-	150	240	-	175	MIL-E-1/1108
1N1206	600	12A	-	150	240	-	175	MIL-E-1/1108
1N1183	50	35A	-	135	900	-	175	MIL-E-1/1135
1N1184	100	35A	-	135	900	-	175	MIL-E-1/1135
1N1185	150	35A	-	135	900	-	175	MIL-E-1/1135
1N1186	200	35A	-	135	900	-	175	MIL-E-1/1135
1N1187	300	35A	-	135	900	-	175	MIL-E-1/1135
1N1188	400	35A	-	135	900	-	175	MIL-E-1/1135
1N1189	500	35A	-	135	900	-	175	MIL-E-1/1135
1N1190	600	35A	-	135	900	-	175	MIL-E-1/1135
1N2172	50	50A	-	150	2000	-	175	MIL-E-1/1196
1N2173	100	50A	-	150	2000	-	175	MIL-E-1/1151
1N2174	200	50A	-	150	2000	-	175	MIL-E-1/1194
1N1682	300	50A	-	150	2000	-	175	MIL-E-1/1195

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Table I - Silicon rectifier diodes (Cont'd.)

JEDEC (EIA) No.	Maximum ratings up to temperature specified				Required derating at higher temp. MA/°C	Maximum oper. temp.		Detailed specification
	V_R 1/	I_F	T_A	T_C		T_A	T_C	
1N1396	50	70A	-	150	2800	-	175	MIL-E-1/1202
1N1397	100	70A	-	150	2800	-	175	MIL-E-1/1202
1N1398	150	70A	-	150	2800	-	175	MIL-E-1/1202
1N1399	200	70A	-	150	2800	-	175	MIL-E-1/1202
1N1400	300	70A	-	150	2800	-	175	MIL-E-1/1202
1N1401	400	70A	-	150	2800	-	175	MIL-E-1/1202
1N1402	500	70A	-	150	2800	-	175	MIL-E-1/1202
1N1403	600	70A	-	150	2800	-	175	MIL-E-1/1202

1/ RMS ac voltage shall not exceed $0.7V_r$.

Symbols: - V_r - Peak reverse voltage across diode, dc volts.

I_F - Average current in forward direction, dc milliamperes (except dc amperes where "A" appears).

T_A - Ambient temperature at diode, degrees centigrade.

T_C - Temperature on or adjacent to mounting surface of the diode, degrees centigrade.

40. SILICON VOLTAGE REFERENCE ELEMENTS

40.1 Silicon voltage reference elements shall be selected in accordance with table II.

Table II - Silicon voltage reference elements

JEDEC (EIA) no.	Design values			Maximum ratings		Detailed specification
	Reference voltage V_{ZR} @ I_{ZB} DC VOLTS		-55°C to 100°C Temperature error ΔV_{ZRT} @ I_{ZB} DC MV	Bias current range 1/ I_{ZB} DC MA	Ambient temperature at reference element	
	min	max				
1N429	5.9	6.5	± 50	7.5 ± 2%	-55 to 100°C	MIL-E-1/1134
1N430	8.0	8.8	± 18	10.0 ± 2%	-55 to 100°C	MIL-E-1/1060

1/ Temperature error ΔV_{ZRT} is not valid for reverse current outside this range.

Symbols: V_{ZR} - Reverse voltage at a specified reverse bias current, I_{ZB}.

ΔV_{ZRT} - Maximum error in V_{ZR} over the specified temperature range at I_{ZB}.

I_{ZB} - Reverse current to which element is biased to obtain design value performance.

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50. VOLTAGE REGULATOR DEVICES

50.1 Voltage regulator devices shall be selected in accordance with table III.

60. TRANSISTORS

60.1 Transistors shall be selected in accordance with table IV:

Table III - Voltage regulator devices

Power dissipation (nominal 25°C value)	Detailed specification
400 mw (3.3 to 10 volts)	MIL-S-19500/127
400 mw (11 to 200 volts)	MIL-S-19500/117
1 watt (6.8 to 200 volts)	MIL-S-19500/115
10 watts (6.8 to 200 volts)	MIL-S-19500/124
50 watts (6.8 to 200 volts)	MIL-S-19500/114

Note: Derate in accordance with the applicable detailed specification indicated above.

Table IV - Transistors

JEDEC (EIA) No.	Type	Gain h_{fe}	Maximum ratings							Detailed specification
			V_{cb} dc Volts	I_c Avg ma	P_c up to temperature specified		Required derating at higher temp. mw/°C	Maximum oper. temp.		
					T_A 65°C	T_C 100°C		T_A	T_C	
2N328A	PNP	18-44	50	50	250	-	2.9	150	-	-
2N329A	PNP	36-88	50	50	250	-	2.9	150	-	-
2N333	NPN	18-40	45	25	100	-	1.2	150	-	MIL-S-19500/37
2N335	NPN	38-91	45	25	100	-	1.2	150	-	MIL-S-19500/37
2N337	NPN	5 MIN	45	20	85	-	1.0	150	-	MIL-S-19500/69
2N338	NPN	10 MIN	45	20	85	-	1.0	150	-	MIL-S-19500/69
2N341	NPN	9-90	125	40	680	-	8.0	150	-	-
2N343	NPN	26-90	60	60	680	-	8.0	150	-	MIL-S-19500/16
2N389	NPN	12-60	1/60	1500	-	48.5w	485	-	150	-
2N424	NPN	12-60	1/80	750	-	48.5w	485	-	150	-
2N491	2/	-	3/60	4/70	300	-	3.6	150	-	-
2N495	PNP	9 MIN	25	50	100	-	1.3	140	-	MIL-S-19500/54
2N496	PNP	1.8 MIN	10	50	100	-	1.3	140	-	MIL-S-19500/85
2N498	NPN	12-36	100	200	-	2.2w	24	-	150	MIL-S-19500/74
2N657	NPN	30-90	100	200	-	2.2w	24	-	150	MIL-S-19500/74

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Table IV - Transistors (Cont'd.)

JEDEC (EIA) No.	Type	Maximum ratings: Gain h_{fe}	V_{cb} dc Volts	I_C Avg ma	P_C up to temperature specified: T_A 65°C	T_C 100°C	Required derating at higher temp. mw/°C	Maximum oper. temp.: T_A	T_C	Detailed specification
2N697	NPN	40-120	1/40	150	-	950	14	-	150	-
2N1026	PNP	18-36	35	100	100	-	1.2	150	-	MIL-S-19500/78
2N1050	NPN	30-90	1/120	500	-	22w	240	-	150	-
2N1132	PNP	2.5 AVG	40	600	-	950	1.4	-	150	-

1/ V_{ce} - Collector-to-emitter voltage, dc volts

2/ - Device has one junction (emitter) contact and two ohmic (base) contacts on n-type crystal

3/ V_{B2E} - Base two-to-emitter voltage, dc volts

4/ I_e - Emitter current, rms milliamperes

Symbols: h_{fe} - Small signal short-circuit forward-transfer current ratio (common emitter base input) - equals Beta
P_C - Collector power dissipation, milliwatts average (watts average where "w" appears)
V_{cb} - Collector voltage, I_C - Collector current
T_A - Ambient temperature at transistor, T_C - Transistor case or body temperature on or adjacent to thermal conduction mounting surface.

APPENDIX III

10. STANDARD MOTOR GENERATOR POWER SUPPLIES

10.1 This appendix lists the standard motor generator power supplies.

20. APPLICABLE DOCUMENTS

20.1 Not applicable.

30. STANDARD MOTOR GENERATOR POWER SUPPLIES

30.1 The standard motor generator power supplies shall be as shown in table I.

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Table I - Standard motor generator power supplies.

Rating						Dimensions (inches) and weights (pounds)												Circuit Breaker				
						Motor Generator								Control panel (max.)								
						Overall (max.)					Mounting											
Kilowatts	Motor voltage (D.C.)	Generator voltage (A.C.)	Generator phases	Generator frequency	Generator power factor	Weight	Length	Width	Height	Frame diameter (max.)	Mounting bolt holes	Bolt hole spacing Length-Width wise		Weight	Depth	Width	Height	Type	Thermal Trip Element Rating (Amperes)	Instant Trip Setting (Amperes)	Ammeter Scale Range (Amperes)	Volt Meter Scale Range (Volts)
1.5	230	120	1	400	0.8	---	--	--	--	----	----	--	--	--	--	--	--	ALB-5	30	----	0-20	150
1.5	115	120	1	400	.8	235	26	15 1/16	15	----	11/16	14 3/4	10 1/2	--	--	--	--	------	--	----	----	---
3	115	120	1	60	.8	375	30	15	15	12-1/4	21/32	14	10	450	18	25	35	ALB-5	35	----	0-30	150
5	230	120	1	60	.8	550	34	18	17	14	21/32	16	11	500	18	25	40	ALB-5	50	----	0-75	150
5	115	120	1	60	.8	550	34	18	17	14	21/32	16	11	650	18	30	40	AQB-250	100G	2350	0-150	150
10	230	120	1	60	.8	1100	42	23	22	18	25/32	18	14	675	18	30	40	AQB-250	100G	2350	0-150	150
10	230	460	3	60	.8	1000	42	23	22	18	25/32	18	14	650	18	30	40	ALB-5	20	----	0-20	600
10	230	450	3	400	.8	1000	42	23	22	18	25/32	18	14	650	18	30	40	ALB-5	20	----	0-20	600
10	115	120	1	60	.8	1100	42	23	22	18	25/32	18	14	725	18	35	40	AQB-250	100G	2350	0-150	150
10	115	450	3	60	.8	1100	42	23	22	18	25/32	18	14	700	18	30	40	ALB-5	20	----	0-20	600
15	230	450	3	60	.8	1350	44	25	24	20	15/16	20	16	675	18	35	40	ALB-5	30	----	0-30	600
15	230	450	3	400	.8	1350	44	25	24	20	15/16	20	16	675	18	35	40	ALB-5	30	----	0-30	600
15	115	450	3	60	.8	1350	44	25	24	20	15/16	20	16	750	18	40	45	ALB-5	30	----	0-30	600
15	115	120	3	60	.8	1305	44	25 1/8	22 3/4	----	15/16	20	16	387	21	11/16	31 7/8	44 1/8	AQB-100	1200-1600	----	0-150	0-150
15	230	120	3	60	.8	1080	38 3/4	24 1/8	21	----	15/16	20	16	677	20		45	46	100A	1200-1600	----	0-200	0-150
30	230	450	3	60	.8	2300	55	28	26	22	1-1/16	25	18	1200	30		40	74	ALB-5	50	----	0-75	600
30	230	450	3	400	.8	2300	55	28	26	22	1-1/16	25	18	1200	30		40	74	ALB-5	50	----	0-80	600
30	115	450	3	60	.8	2500	55	31	29	25	1-1/16	25	20	1300	30		45	74	ALB-5	50	----	0-75	600
30	230	120	3	60	.8	4125	79 1/2	27	26 3/4	1-1/16	1-1/16	30 1/4	25	550	22	1/4	28	62	------	--	----	----	---
60	230	450	3	60	.8	3600	63	34	33	28	1-5/16	30	24	1500	30		50	74	AQB-250	100G	2350	0-150	600
100	230	450	3	60	.8	5000	67	36	35	30	1-5/16	34	24	1500	30		50	74	AQB-250	160G	3900	0-200	600

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